ArtExtract_1

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1 ArtExtract (Model for Art classification)

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This notebook implements the pieces of work as demanded in the Task 1

Due to compute constraints, I preferred to proceed with a smaller dataset : Wikiart-tiny from huggingface which is the subset of the original wikiart dataset

[ ]:
```
# importing the wikiart-tiny dataset
from datasets import load_dataset
dataset = load_dataset("yonigozlan/wikiart-tiny", split='train')
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94:
UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab
(https://huggingface.co/settings/tokens), set it as secret in your Google Colab
and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access
public models or datasets.
  warnings.warn(
```

```
README.md: 0.00B [00:00, ?B/s]
```

```
data/train-00000-of-00001.parquet:   0%|          | 0.00/459M [00:00<?, ?B/s]
```

```
Generating train split:   0%|          | 0/1000 [00:00<?, ? examples/s]
```

[ ]:
```
dataset
```

[ ]:
```
Dataset({
    features: ['image', 'artist', 'genre', 'style'],
    num_rows: 1000
})
```

[ ]:
```
dataset[0]['image'].size, dataset[1]['image'].size
```

[ ]:
```
((750, 597), (1382, 1659))
```

[ ]:
```
## we can see clearly that the resolution of every image is different
## so we will resize the dataset images to same resolution
IMAGE_SIZE = [224, 224]
from PIL import Image
def resize_images(examples):
  resized_images = [image.convert("RGB").resize(IMAGE_SIZE) for image in␣
 ↪examples['image']]
  examples['image'] = resized_images
  return examples

## apply the transformation
dataset = dataset.map(resize_images, batched=True)
```

```
Map:   0%|          | 0/1000 [00:00<?, ? examples/s]
```

[ ]:
```
# now check whether the images are resized
dataset[1]['image'].size
```

[ ]: (224, 224)

2 My approach :

1. Use CNN for classification
2. Use CRNN for classifcation
   - LSTM v/s
   - GRU
3. Compare results of all three architectures

2.0.1 Data processing

[ ]:
```
import numpy as np

# now, sequentially mapping the class ids as zero indexed as the loss function␣
 ↪demands

artist_classes = sorted(dataset.unique('artist'))
genre_classes = sorted(dataset.unique('genre'))
style_classes = sorted(dataset.unique('style'))

artist_classes_num = len(artist_classes)
genre_classes_num = len(genre_classes)
style_classes_num = len(style_classes)

artist_to_id = {v: i for i, v in enumerate(artist_classes)}
genre_to_id = {v: i for i, v in enumerate(genre_classes)}
style_to_id = {v: i for i, v in enumerate(style_classes)}
```

```
def preprocess_data(examples):

  # storing the artist ids in new lists :
  examples['artist_labels'] = np.array([int(artist_to_id[artist_id]) for␣
 ↪artist_id in examples['artist']], dtype=np.int32)
  examples['genre_labels'] = np.array([int(genre_to_id[genre_id]) for genre_id␣
 ↪in examples['genre']], dtype=np.int32)
  examples['style_labels'] = np.array([int(style_to_id[style_id]) for style_id␣
 ↪in examples['style']], dtype=np.int32)

  return examples
```

[ ]:
```
# now we will apply the preprocess_data function
# also, we need to convert the processed data into tf.data.Dataset objects for␣
 ↪efficient training

import tensorflow as tf

# process the complete data
full_data_processed = dataset.map(preprocess_data, batched=True)

# now split the data into train, val and test
full_data_processed = full_data_processed.train_test_split(test_size=0.2)
train_data = full_data_processed['train'] # 80%
test_data = full_data_processed['test']   # 20%
test_data = test_data.train_test_split(test_size=0.5)
val_data = test_data['train']             # 10%
test_data = test_data['test']             # 10%

def preprocess_tf(image, labels):
    image = tf.cast(image, tf.float32) / 255.0
    labels = (labels['artist_labels'], labels['genre_labels'],␣
 ↪labels['style_labels'])
    return image, labels

# now covert it to tensorflow data
train_data_tf = train_data.to_tf_dataset(
    columns=["image"],
    label_cols=["artist_labels", "genre_labels", "style_labels"],
    shuffle=True, # shuffle for train
    batch_size=32
).map(preprocess_tf, num_parallel_calls=tf.data.AUTOTUNE).prefetch(tf.data.
 ↪AUTOTUNE)

test_data_tf = test_data.to_tf_dataset(
    columns=["image"],
```

```
    label_cols=["artist_labels", "genre_labels", "style_labels"],
    shuffle=False,
    batch_size=32
).map(preprocess_tf, num_parallel_calls=tf.data.AUTOTUNE).prefetch(tf.data.
 ↪AUTOTUNE)

val_data_tf = val_data.to_tf_dataset(
    columns=["image"],
    label_cols=["artist_labels", "genre_labels", "style_labels"],
    shuffle=False,
    batch_size=32
).map(preprocess_tf, num_parallel_calls=tf.data.AUTOTUNE).prefetch(tf.data.
 ↪AUTOTUNE)
```

```
Map:   0%|          | 0/1000 [00:00<?, ? examples/s]
```

```
/usr/local/lib/python3.12/dist-packages/datasets/arrow_dataset.py:403:
FutureWarning: The output of `to_tf_dataset` will change when a passing single
element list for `labels` or `columns` in the next datasets version. To return a
tuple structure rather than dict, pass a single string.
Old behaviour: columns=['a'], labels=['labels'] -> (tf.Tensor, tf.Tensor)
             : columns='a', labels='labels' -> (tf.Tensor, tf.Tensor)
New behaviour: columns=['a'],labels=['labels'] -> ({'a': tf.Tensor}, {'labels':
tf.Tensor})
             : columns='a', labels='labels' -> (tf.Tensor, tf.Tensor)
  warnings.warn(
```

2.1 Starting with the CNN model

I considered the following CNN architectures : - LeNet - VGGnet - ResNet - AlexNet - EfficientNet - DenseNet

I chose to proceed with ResNet50 as : - The skip connections allows the training of such deep models without performance degradation, this is a crucial advantage over shallow VGG, Alex and LeNet architectures - The 50 deep layers are essential for differentiating subtle artistic classes, genre and artist in a diverse dataset like wikiart - suitability for transfer learning, allows to load pretrained imagenet weights instead of training the complete model from scratch avoiding overfitting.

[ ]:
```
# First make a ResNet 50 model for image classification
# make the model architectutre, remove the classification layer and freeze␣
 ↪trainable layers
from tensorflow.keras.layers import Flatten, Dense
from tensorflow.keras.models import Model
from tensorflow.keras.applications.resnet50 import ResNet50
from tensorflow.keras.layers import GlobalAveragePooling2D

myResnet = ResNet50(weights='imagenet', input_shape=IMAGE_SIZE + [3],␣
 ↪include_top=False) ## remove the fully connected classificaiton layer
```

```python
for layer in myResnet.layers:
  layer.trainable = False

# make the flatten layer
flattenLayer = GlobalAveragePooling2D()(myResnet.output)

# make the classification heads

# 1. artist classification
artist_output = Dense(artist_classes_num, activation='softmax', name␣
↪='artist_output')(flattenLayer)

# 2. genre classification
genre_output = Dense(genre_classes_num, activation='softmax', name␣
↪='genre_output')(flattenLayer)

# 3. style classification
style_output = Dense(style_classes_num, activation='softmax', name␣
↪='style_output')(flattenLayer)

# create the multi-output model
model = Model(inputs=myResnet.input, outputs=[artist_output, genre_output,␣
↪style_output])

print(model.summary())
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-
applications/resnet/resnet50_weights_tf_dim_ordering_tf_kernels_notop.h5
94765736/94765736              5s
0us/step
```

Model: "functional"

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 224, 224, 3)	0	-
conv1_pad (ZeroPadding2D)	(None, 230, 230, 3)	0	input_layer[0][0]
conv1_conv (Conv2D)	(None, 112, 112, 64)	9,472	conv1_pad[0][0]
conv1_bn (BatchNormalizatio…	(None, 112, 112, 64)	256	conv1_conv[0][0]

conv1_relu (Activation)	(None, 112, 112, 64)	0	conv1_bn[0][0]
pool1_pad (ZeroPadding2D)	(None, 114, 114, 64)	0	conv1_relu[0][0]
pool1_pool (MaxPooling2D)	(None, 56, 56, 64)	0	pool1_pad[0][0]
conv2_block1_1_conv (Conv2D)	(None, 56, 56, 64)	4,160	pool1_pool[0][0]
conv2_block1_1_bn (BatchNormalizatio…	(None, 56, 56, 64)	256	conv2_block1_1_c…
conv2_block1_1_relu (Activation)	(None, 56, 56, 64)	0	conv2_block1_1_b…
conv2_block1_2_conv (Conv2D)	(None, 56, 56, 64)	36,928	conv2_block1_1_r…
conv2_block1_2_bn (BatchNormalizatio…	(None, 56, 56, 64)	256	conv2_block1_2_c…
conv2_block1_2_relu (Activation)	(None, 56, 56, 64)	0	conv2_block1_2_b…
conv2_block1_0_conv (Conv2D)	(None, 56, 56, 256)	16,640	pool1_pool[0][0]
conv2_block1_3_conv (Conv2D)	(None, 56, 56, 256)	16,640	conv2_block1_2_r…
conv2_block1_0_bn (BatchNormalizatio…	(None, 56, 56, 256)	1,024	conv2_block1_0_c…
conv2_block1_3_bn (BatchNormalizatio…	(None, 56, 56, 256)	1,024	conv2_block1_3_c…
conv2_block1_add (Add)	(None, 56, 56, 256)	0	conv2_block1_0_b… conv2_block1_3_b…
conv2_block1_out (Activation)	(None, 56, 56, 256)	0	conv2_block1_add…
conv2_block2_1_conv (Conv2D)	(None, 56, 56, 64)	16,448	conv2_block1_out…

conv2_block2_1_bn (BatchNormalizatio…	(None, 56, 56, 64)	256	conv2_block2_1_c…
conv2_block2_1_relu (Activation)	(None, 56, 56, 64)	0	conv2_block2_1_b…
conv2_block2_2_conv (Conv2D)	(None, 56, 56, 64)	36,928	conv2_block2_1_r…
conv2_block2_2_bn (BatchNormalizatio…	(None, 56, 56, 64)	256	conv2_block2_2_c…
conv2_block2_2_relu (Activation)	(None, 56, 56, 64)	0	conv2_block2_2_b…
conv2_block2_3_conv (Conv2D)	(None, 56, 56, 256)	16,640	conv2_block2_2_r…
conv2_block2_3_bn (BatchNormalizatio…	(None, 56, 56, 256)	1,024	conv2_block2_3_c…
conv2_block2_add (Add)	(None, 56, 56, 256)	0	conv2_block1_out… conv2_block2_3_b…
conv2_block2_out (Activation)	(None, 56, 56, 256)	0	conv2_block2_add…
conv2_block3_1_conv (Conv2D)	(None, 56, 56, 64)	16,448	conv2_block2_out…
conv2_block3_1_bn (BatchNormalizatio…	(None, 56, 56, 64)	256	conv2_block3_1_c…
conv2_block3_1_relu (Activation)	(None, 56, 56, 64)	0	conv2_block3_1_b…
conv2_block3_2_conv (Conv2D)	(None, 56, 56, 64)	36,928	conv2_block3_1_r…
conv2_block3_2_bn (BatchNormalizatio…	(None, 56, 56, 64)	256	conv2_block3_2_c…
conv2_block3_2_relu (Activation)	(None, 56, 56, 64)	0	conv2_block3_2_b…
conv2_block3_3_conv (Conv2D)	(None, 56, 56, 256)	16,640	conv2_block3_2_r…

conv2_block3_3_bn (BatchNormalizatio…	(None, 56, 56, 256)	1,024	conv2_block3_3_c…
conv2_block3_add (Add)	(None, 56, 56, 256)	0	conv2_block2_out… conv2_block3_3_b…
conv2_block3_out (Activation)	(None, 56, 56, 256)	0	conv2_block3_add…
conv3_block1_1_conv (Conv2D)	(None, 28, 28, 128)	32,896	conv2_block3_out…
conv3_block1_1_bn (BatchNormalizatio…	(None, 28, 28, 128)	512	conv3_block1_1_c…
conv3_block1_1_relu (Activation)	(None, 28, 28, 128)	0	conv3_block1_1_b…
conv3_block1_2_conv (Conv2D)	(None, 28, 28, 128)	147,584	conv3_block1_1_r…
conv3_block1_2_bn (BatchNormalizatio…	(None, 28, 28, 128)	512	conv3_block1_2_c…
conv3_block1_2_relu (Activation)	(None, 28, 28, 128)	0	conv3_block1_2_b…
conv3_block1_0_conv (Conv2D)	(None, 28, 28, 512)	131,584	conv2_block3_out…
conv3_block1_3_conv (Conv2D)	(None, 28, 28, 512)	66,048	conv3_block1_2_r…
conv3_block1_0_bn (BatchNormalizatio…	(None, 28, 28, 512)	2,048	conv3_block1_0_c…
conv3_block1_3_bn (BatchNormalizatio…	(None, 28, 28, 512)	2,048	conv3_block1_3_c…
conv3_block1_add (Add)	(None, 28, 28, 512)	0	conv3_block1_0_b… conv3_block1_3_b…
conv3_block1_out (Activation)	(None, 28, 28, 512)	0	conv3_block1_add…
conv3_block2_1_conv (Conv2D)	(None, 28, 28, 128)	65,664	conv3_block1_out…

conv3_block2_1_bn (BatchNormalizatio…	(None, 28, 28, 128)	512	conv3_block2_1_c…
conv3_block2_1_relu (Activation)	(None, 28, 28, 128)	0	conv3_block2_1_b…
conv3_block2_2_conv (Conv2D)	(None, 28, 28, 128)	147,584	conv3_block2_1_r…
conv3_block2_2_bn (BatchNormalizatio…	(None, 28, 28, 128)	512	conv3_block2_2_c…
conv3_block2_2_relu (Activation)	(None, 28, 28, 128)	0	conv3_block2_2_b…
conv3_block2_3_conv (Conv2D)	(None, 28, 28, 512)	66,048	conv3_block2_2_r…
conv3_block2_3_bn (BatchNormalizatio…	(None, 28, 28, 512)	2,048	conv3_block2_3_c…
conv3_block2_add (Add)	(None, 28, 28, 512)	0	conv3_block1_out… conv3_block2_3_b…
conv3_block2_out (Activation)	(None, 28, 28, 512)	0	conv3_block2_add…
conv3_block3_1_conv (Conv2D)	(None, 28, 28, 128)	65,664	conv3_block2_out…
conv3_block3_1_bn (BatchNormalizatio…	(None, 28, 28, 128)	512	conv3_block3_1_c…
conv3_block3_1_relu (Activation)	(None, 28, 28, 128)	0	conv3_block3_1_b…
conv3_block3_2_conv (Conv2D)	(None, 28, 28, 128)	147,584	conv3_block3_1_r…
conv3_block3_2_bn (BatchNormalizatio…	(None, 28, 28, 128)	512	conv3_block3_2_c…
conv3_block3_2_relu (Activation)	(None, 28, 28, 128)	0	conv3_block3_2_b…
conv3_block3_3_conv (Conv2D)	(None, 28, 28, 512)	66,048	conv3_block3_2_r…

conv3_block3_3_bn (BatchNormalizatio…	(None, 28, 28, 512)	2,048	conv3_block3_3_c…
conv3_block3_add (Add)	(None, 28, 28, 512)	0	conv3_block2_out… conv3_block3_3_b…
conv3_block3_out (Activation)	(None, 28, 28, 512)	0	conv3_block3_add…
conv3_block4_1_conv (Conv2D)	(None, 28, 28, 128)	65,664	conv3_block3_out…
conv3_block4_1_bn (BatchNormalizatio…	(None, 28, 28, 128)	512	conv3_block4_1_c…
conv3_block4_1_relu (Activation)	(None, 28, 28, 128)	0	conv3_block4_1_b…
conv3_block4_2_conv (Conv2D)	(None, 28, 28, 128)	147,584	conv3_block4_1_r…
conv3_block4_2_bn (BatchNormalizatio…	(None, 28, 28, 128)	512	conv3_block4_2_c…
conv3_block4_2_relu (Activation)	(None, 28, 28, 128)	0	conv3_block4_2_b…
conv3_block4_3_conv (Conv2D)	(None, 28, 28, 512)	66,048	conv3_block4_2_r…
conv3_block4_3_bn (BatchNormalizatio…	(None, 28, 28, 512)	2,048	conv3_block4_3_c…
conv3_block4_add (Add)	(None, 28, 28, 512)	0	conv3_block3_out… conv3_block4_3_b…
conv3_block4_out (Activation)	(None, 28, 28, 512)	0	conv3_block4_add…
conv4_block1_1_conv (Conv2D)	(None, 14, 14, 256)	131,328	conv3_block4_out…
conv4_block1_1_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block1_1_c…
conv4_block1_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block1_1_b…

conv4_block1_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block1_1_r…
conv4_block1_2_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block1_2_c…
conv4_block1_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block1_2_b…
conv4_block1_0_conv (Conv2D)	(None, 14, 14, 1024)	525,312	conv3_block4_out…
conv4_block1_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block1_2_r…
conv4_block1_0_bn (BatchNormalizatio…	(None, 14, 14, 1024)	4,096	conv4_block1_0_c…
conv4_block1_3_bn (BatchNormalizatio…	(None, 14, 14, 1024)	4,096	conv4_block1_3_c…
conv4_block1_add (Add)	(None, 14, 14, 1024)	0	conv4_block1_0_b… conv4_block1_3_b…
conv4_block1_out (Activation)	(None, 14, 14, 1024)	0	conv4_block1_add…
conv4_block2_1_conv (Conv2D)	(None, 14, 14, 256)	262,400	conv4_block1_out…
conv4_block2_1_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block2_1_c…
conv4_block2_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block2_1_b…
conv4_block2_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block2_1_r…
conv4_block2_2_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block2_2_c…
conv4_block2_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block2_2_b…
conv4_block2_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block2_2_r…

conv4_block2_3_bn (BatchNormalizatio…	(None, 14, 14, 1024)	4,096	conv4_block2_3_c…
conv4_block2_add (Add)	(None, 14, 14, 1024)	0	conv4_block1_out… conv4_block2_3_b…
conv4_block2_out (Activation)	(None, 14, 14, 1024)	0	conv4_block2_add…
conv4_block3_1_conv (Conv2D)	(None, 14, 14, 256)	262,400	conv4_block2_out…
conv4_block3_1_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block3_1_c…
conv4_block3_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block3_1_b…
conv4_block3_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block3_1_r…
conv4_block3_2_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block3_2_c…
conv4_block3_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block3_2_b…
conv4_block3_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block3_2_r…
conv4_block3_3_bn (BatchNormalizatio…	(None, 14, 14, 1024)	4,096	conv4_block3_3_c…
conv4_block3_add (Add)	(None, 14, 14, 1024)	0	conv4_block2_out… conv4_block3_3_b…
conv4_block3_out (Activation)	(None, 14, 14, 1024)	0	conv4_block3_add…
conv4_block4_1_conv (Conv2D)	(None, 14, 14, 256)	262,400	conv4_block3_out…
conv4_block4_1_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block4_1_c…
conv4_block4_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block4_1_b…

conv4_block4_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block4_1_r…
conv4_block4_2_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block4_2_c…
conv4_block4_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block4_2_b…
conv4_block4_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block4_2_r…
conv4_block4_3_bn (BatchNormalizatio…	(None, 14, 14, 1024)	4,096	conv4_block4_3_c…
conv4_block4_add (Add)	(None, 14, 14, 1024)	0	conv4_block3_out… conv4_block4_3_b…
conv4_block4_out (Activation)	(None, 14, 14, 1024)	0	conv4_block4_add…
conv4_block5_1_conv (Conv2D)	(None, 14, 14, 256)	262,400	conv4_block4_out…
conv4_block5_1_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block5_1_c…
conv4_block5_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block5_1_b…
conv4_block5_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block5_1_r…
conv4_block5_2_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block5_2_c…
conv4_block5_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block5_2_b…
conv4_block5_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block5_2_r…
conv4_block5_3_bn (BatchNormalizatio…	(None, 14, 14, 1024)	4,096	conv4_block5_3_c…
conv4_block5_add (Add)	(None, 14, 14, 1024)	0	conv4_block4_out… conv4_block5_3_b…

conv4_block5_out (Activation)	(None, 14, 14, 1024)	0	conv4_block5_add…
conv4_block6_1_conv (Conv2D)	(None, 14, 14, 256)	262,400	conv4_block5_out…
conv4_block6_1_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block6_1_c…
conv4_block6_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block6_1_b…
conv4_block6_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block6_1_r…
conv4_block6_2_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block6_2_c…
conv4_block6_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block6_2_b…
conv4_block6_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block6_2_r…
conv4_block6_3_bn (BatchNormalizatio…	(None, 14, 14, 1024)	4,096	conv4_block6_3_c…
conv4_block6_add (Add)	(None, 14, 14, 1024)	0	conv4_block5_out… conv4_block6_3_b…
conv4_block6_out (Activation)	(None, 14, 14, 1024)	0	conv4_block6_add…
conv5_block1_1_conv (Conv2D)	(None, 7, 7, 512)	524,800	conv4_block6_out…
conv5_block1_1_bn (BatchNormalizatio…	(None, 7, 7, 512)	2,048	conv5_block1_1_c…
conv5_block1_1_relu (Activation)	(None, 7, 7, 512)	0	conv5_block1_1_b…
conv5_block1_2_conv (Conv2D)	(None, 7, 7, 512)	2,359,808	conv5_block1_1_r…
conv5_block1_2_bn (BatchNormalizatio…	(None, 7, 7, 512)	2,048	conv5_block1_2_c…

conv5_block1_2_relu (Activation)	(None, 7, 7, 512)	0	conv5_block1_2_b…
conv5_block1_0_conv (Conv2D)	(None, 7, 7, 2048)	2,099,200	conv4_block6_out…
conv5_block1_3_conv (Conv2D)	(None, 7, 7, 2048)	1,050,624	conv5_block1_2_r…
conv5_block1_0_bn (BatchNormalizatio…	(None, 7, 7, 2048)	8,192	conv5_block1_0_c…
conv5_block1_3_bn (BatchNormalizatio…	(None, 7, 7, 2048)	8,192	conv5_block1_3_c…
conv5_block1_add (Add)	(None, 7, 7, 2048)	0	conv5_block1_0_b… conv5_block1_3_b…
conv5_block1_out (Activation)	(None, 7, 7, 2048)	0	conv5_block1_add…
conv5_block2_1_conv (Conv2D)	(None, 7, 7, 512)	1,049,088	conv5_block1_out…
conv5_block2_1_bn (BatchNormalizatio…	(None, 7, 7, 512)	2,048	conv5_block2_1_c…
conv5_block2_1_relu (Activation)	(None, 7, 7, 512)	0	conv5_block2_1_b…
conv5_block2_2_conv (Conv2D)	(None, 7, 7, 512)	2,359,808	conv5_block2_1_r…
conv5_block2_2_bn (BatchNormalizatio…	(None, 7, 7, 512)	2,048	conv5_block2_2_c…
conv5_block2_2_relu (Activation)	(None, 7, 7, 512)	0	conv5_block2_2_b…
conv5_block2_3_conv (Conv2D)	(None, 7, 7, 2048)	1,050,624	conv5_block2_2_r…
conv5_block2_3_bn (BatchNormalizatio…	(None, 7, 7, 2048)	8,192	conv5_block2_3_c…
conv5_block2_add (Add)	(None, 7, 7, 2048)	0	conv5_block1_out… conv5_block2_3_b…

conv5_block2_out (Activation)	(None, 7, 7, 2048)	0	conv5_block2_add…
conv5_block3_1_conv (Conv2D)	(None, 7, 7, 512)	1,049,088	conv5_block2_out…
conv5_block3_1_bn (BatchNormalizatio…	(None, 7, 7, 512)	2,048	conv5_block3_1_c…
conv5_block3_1_relu (Activation)	(None, 7, 7, 512)	0	conv5_block3_1_b…
conv5_block3_2_conv (Conv2D)	(None, 7, 7, 512)	2,359,808	conv5_block3_1_r…
conv5_block3_2_bn (BatchNormalizatio…	(None, 7, 7, 512)	2,048	conv5_block3_2_c…
conv5_block3_2_relu (Activation)	(None, 7, 7, 512)	0	conv5_block3_2_b…
conv5_block3_3_conv (Conv2D)	(None, 7, 7, 2048)	1,050,624	conv5_block3_2_r…
conv5_block3_3_bn (BatchNormalizatio…	(None, 7, 7, 2048)	8,192	conv5_block3_3_c…
conv5_block3_add (Add)	(None, 7, 7, 2048)	0	conv5_block2_out… conv5_block3_3_b…
conv5_block3_out (Activation)	(None, 7, 7, 2048)	0	conv5_block3_add…
global_average_poo… (GlobalAveragePool…	(None, 2048)	0	conv5_block3_out…
artist_output (Dense)	(None, 24)	49,176	global_average_p…
genre_output (Dense)	(None, 10)	20,490	global_average_p…
style_output (Dense)	(None, 16)	32,784	global_average_p…

```
Total params: 23,690,162 (90.37 MB)

Trainable params: 102,450 (400.20 KB)

Non-trainable params: 23,587,712 (89.98 MB)

None
```

2.2 The Hyperparameters

2.2.1 Activation function

While deciding which activation function to use, i had choices for :

- Sigmoid
- ReLU
- Softmax
- Swish
- TanH

I landed on going with softmax as it fits the best in multiclass classification as it gives the probabilities for each class in artist, genre and style category that sums up to 1.

Reference : Activation functions for Multiclass Classification

2.2.2 Optimizer

Choices I came upon : - Adam - SGD

Went with Adam. Even adam had its own disadvantage of converging at the very beginning of the training, in our case, we have freezed the maximum training layers of the resnet50 since we are using imagenet weights, so we only have the classification heads to be trained. So, here using adam is no harm and will work best

Reference : Overview of optimizers for DNN

2.2.3 Loss Function

categorical_crossentropy is to be used but since, it requires labels in the form of one-hot encoded vectors and our dataset has it as integer IDs, it's better to go with sparse_categorical_crossentropy

Reference : Which loss function and metrics to use for multi-class classification

[ ]:
```
from tensorflow.keras.metrics import F1Score

metrics ={
    "artist_output": ["accuracy"],
    "genre_output": ["accuracy"],
    "style_output": ["accuracy"]
}
```

```
optimizer = tf.keras.optimizers.Adam(learning_rate=1e-4)

model.compile(
    optimizer=optimizer,
    loss={
        'artist_output': 'sparse_categorical_crossentropy',
        'genre_output': 'sparse_categorical_crossentropy',
        'style_output': 'sparse_categorical_crossentropy'
    },
    metrics = metrics
)

print("Model compiled successfully!")
```

```
Model compiled successfully!
```

[ ]:
```
history = model.fit(
    train_data_tf,
    validation_data=val_data_tf,
    epochs=10
)
```

```
Epoch 1/10
25/25              40s 760ms/step -
artist_output_accuracy: 0.0211 - artist_output_loss: 3.7174 -
genre_output_accuracy: 0.2375 - genre_output_loss: 2.4130 - loss: 8.9020 -
style_output_accuracy: 0.0988 - style_output_loss: 2.7715 -
val_artist_output_accuracy: 0.0400 - val_artist_output_loss: 3.4339 -
val_genre_output_accuracy: 0.1600 - val_genre_output_loss: 2.4403 - val_loss:
8.2023 - val_style_output_accuracy: 0.1100 - val_style_output_loss: 2.4622
Epoch 2/10
25/25              6s 243ms/step -
artist_output_accuracy: 0.0288 - artist_output_loss: 3.3559 -
genre_output_accuracy: 0.2407 - genre_output_loss: 2.2649 - loss: 8.0515 -
style_output_accuracy: 0.1707 - style_output_loss: 2.4308 -
val_artist_output_accuracy: 0.0400 - val_artist_output_loss: 3.2585 -
val_genre_output_accuracy: 0.1600 - val_genre_output_loss: 2.2827 - val_loss:
7.6837 - val_style_output_accuracy: 0.2200 - val_style_output_loss: 2.2544
Epoch 3/10
25/25              7s 260ms/step -
artist_output_accuracy: 0.0226 - artist_output_loss: 3.1806 -
genre_output_accuracy: 0.2201 - genre_output_loss: 2.1700 - loss: 7.6153 -
style_output_accuracy: 0.2184 - style_output_loss: 2.2648 -
val_artist_output_accuracy: 0.0800 - val_artist_output_loss: 3.1922 -
val_genre_output_accuracy: 0.1600 - val_genre_output_loss: 2.2215 - val_loss:
7.4698 - val_style_output_accuracy: 0.1900 - val_style_output_loss: 2.1769
Epoch 4/10
25/25              6s 220ms/step -
```

```
artist_output_accuracy: 0.0534 - artist_output_loss: 3.1083 -
genre_output_accuracy: 0.2283 - genre_output_loss: 2.1420 - loss: 7.4935 -
style_output_accuracy: 0.2573 - style_output_loss: 2.2432 -
val_artist_output_accuracy: 0.0900 - val_artist_output_loss: 3.1646 -
val_genre_output_accuracy: 0.1600 - val_genre_output_loss: 2.1956 - val_loss:
7.3785 - val_style_output_accuracy: 0.3700 - val_style_output_loss: 2.1350
Epoch 5/10
25/25            6s 243ms/step -
artist_output_accuracy: 0.1098 - artist_output_loss: 3.0673 -
genre_output_accuracy: 0.2340 - genre_output_loss: 2.1318 - loss: 7.3892 -
style_output_accuracy: 0.2873 - style_output_loss: 2.1901 -
val_artist_output_accuracy: 0.0900 - val_artist_output_loss: 3.1528 -
val_genre_output_accuracy: 0.1600 - val_genre_output_loss: 2.1860 - val_loss:
7.3389 - val_style_output_accuracy: 0.3600 - val_style_output_loss: 2.1173
Epoch 6/10
25/25            10s 243ms/step -
artist_output_accuracy: 0.1290 - artist_output_loss: 3.0236 -
genre_output_accuracy: 0.2249 - genre_output_loss: 2.1123 - loss: 7.2482 -
style_output_accuracy: 0.3022 - style_output_loss: 2.1123 -
val_artist_output_accuracy: 0.1100 - val_artist_output_loss: 3.1421 -
val_genre_output_accuracy: 0.1600 - val_genre_output_loss: 2.1777 - val_loss:
7.3126 - val_style_output_accuracy: 0.3600 - val_style_output_loss: 2.1031
Epoch 7/10
25/25            6s 248ms/step -
artist_output_accuracy: 0.1261 - artist_output_loss: 3.0359 -
genre_output_accuracy: 0.2237 - genre_output_loss: 2.1393 - loss: 7.2977 -
style_output_accuracy: 0.3065 - style_output_loss: 2.1226 -
val_artist_output_accuracy: 0.1500 - val_artist_output_loss: 3.1355 -
val_genre_output_accuracy: 0.1600 - val_genre_output_loss: 2.1731 - val_loss:
7.2923 - val_style_output_accuracy: 0.3600 - val_style_output_loss: 2.0980
Epoch 8/10
25/25            6s 231ms/step -
artist_output_accuracy: 0.1418 - artist_output_loss: 3.0132 -
genre_output_accuracy: 0.2242 - genre_output_loss: 2.1290 - loss: 7.3621 -
style_output_accuracy: 0.2616 - style_output_loss: 2.2199 -
val_artist_output_accuracy: 0.1100 - val_artist_output_loss: 3.1270 -
val_genre_output_accuracy: 0.1600 - val_genre_output_loss: 2.1720 - val_loss:
7.2824 - val_style_output_accuracy: 0.3600 - val_style_output_loss: 2.0841
Epoch 9/10
25/25            6s 230ms/step -
artist_output_accuracy: 0.1377 - artist_output_loss: 3.0356 -
genre_output_accuracy: 0.2349 - genre_output_loss: 2.0988 - loss: 7.2827 -
style_output_accuracy: 0.2706 - style_output_loss: 2.1483 -
val_artist_output_accuracy: 0.1500 - val_artist_output_loss: 3.1109 -
val_genre_output_accuracy: 0.1700 - val_genre_output_loss: 2.1654 - val_loss:
7.2533 - val_style_output_accuracy: 0.3500 - val_style_output_loss: 2.0743
Epoch 10/10
25/25            6s 229ms/step -
```

```
artist_output_accuracy: 0.1809 - artist_output_loss: 2.9797 -
genre_output_accuracy: 0.2503 - genre_output_loss: 2.0931 - loss: 7.1804 -
style_output_accuracy: 0.2796 - style_output_loss: 2.1076 -
val_artist_output_accuracy: 0.1500 - val_artist_output_loss: 3.1138 -
val_genre_output_accuracy: 0.1600 - val_genre_output_loss: 2.1641 - val_loss:
7.2522 - val_style_output_accuracy: 0.3500 - val_style_output_loss: 2.0694
```

[ ]:
```
results = model.evaluate(test_data_tf, return_dict=True)
for k, v in results.items():
    print(f"{k}: {v:.4f}")
```

```
4/4              1s 152ms/step -
artist_output_accuracy: 0.1204 - artist_output_loss: 3.0539 -
genre_output_accuracy: 0.2674 - genre_output_loss: 2.0773 - loss: 7.3420 -
style_output_accuracy: 0.3445 - style_output_loss: 2.2287
artist_output_accuracy: 0.0900
artist_output_loss: 3.0834
genre_output_accuracy: 0.2700
genre_output_loss: 2.0511
loss: 7.3813
style_output_accuracy: 0.3300
style_output_loss: 2.2918
```

Here, the results are very low than what is expected.

But considering that the dataset is very small (trained on only 800 images and validated and tested on 100, 100), the aim of this task is to get onto the CRNN model and study the difference it makes in the results which have been taken care of in the next cells :

2.3 Classification Metrics

Relying on accuracy is not worth in multiclass classifications

Choices : - Macro F1 - Balanced accuracy

Since our data is unbalanced, maro f1 is preferred

[ ]:
```
# calculating f1_scores
y_pred = model.predict(test_data_tf)
y_pred_classes = [np.argmax(pred, axis=1) for pred in y_pred]
y_true = [[], [], []]

for x, y in test_data_tf:
    for i in range(3):
        y_true[i].append(y[i])

y_true = [tf.concat(part, axis=0) for part in y_true]
y_true_np = [y.numpy() for y in y_true]
```

```
4/4              9s 1s/step
```

2.3.1 F1 Score for CNN

[ ]:
```
from sklearn.metrics import f1_score
f1_scores = [
    f1_score(y_true_np[i], y_pred_classes[i], average='macro')
    for i in range(3)
]

overall_f1 = sum(f1_scores) / len(f1_scores)
print("Overall Macro F1 for CNN:", overall_f1)
```

```
Overall Macro F1: 0.03668732783313646
```

2.3.2 Predicted Class Distribution for CNN model

[ ]:
```
for i in range(3):
    print(f"\nOutput {i+1}")
    print("True distribution:", np.bincount(y_true_np[i]))
    print("Pred distribution:", np.bincount(y_pred_classes[i]))
```

```
Output 1
True distribution: [ 1  4  5  4 10  3  1  3  3  0  1  3  3  4  9  2  7  7  5  3
5  3  8  6]
Pred distribution: [ 0  0  0  0  0  0  0  0  0  0  0  0  0  0 42  0  0  0  0  0
0  0 58]

Output 2
True distribution: [ 9 14  2 27  2 20  6  4  4 12]
Pred distribution: [ 0  0  0 97  0  3]

Output 3
True distribution: [ 0  0  8  5  5  3  1 32  2  7  0 14 10  4  9]
Pred distribution: [ 0  0  0  0  0  0  0 95  0  0  0  0  5]
```

Here it is obseravable that the imbalance is much high in the predictions for a few classses

2.3.3 Classification Report for CNN

[ ]:
```
from sklearn.metrics import classification_report

for i in range(3):
    print(f"\nOutput {i+1}")
    print(classification_report(y_true_np[i], y_pred_classes[i]))
```

```
Output 1
              precision    recall  f1-score   support
```

0	0.00	0.00	0.00	1
1	0.00	0.00	0.00	4
2	0.00	0.00	0.00	5
3	0.00	0.00	0.00	4
4	0.00	0.00	0.00	10
5	0.00	0.00	0.00	3
6	0.00	0.00	0.00	1
7	0.00	0.00	0.00	3
8	0.00	0.00	0.00	3
10	0.00	0.00	0.00	1
11	0.00	0.00	0.00	3
12	0.00	0.00	0.00	3
13	0.00	0.00	0.00	4
14	0.10	0.44	0.16	9
15	0.00	0.00	0.00	2
16	0.00	0.00	0.00	7
17	0.00	0.00	0.00	7
18	0.00	0.00	0.00	5
19	0.00	0.00	0.00	3
20	0.00	0.00	0.00	5
21	0.00	0.00	0.00	3
22	0.09	0.62	0.15	8
23	0.00	0.00	0.00	6
accuracy			0.09	100
macro avg	0.01	0.05	0.01	100
weighted avg	0.02	0.09	0.03	100

Output 2

	precision	recall	f1-score	support
0	0.00	0.00	0.00	9
1	0.00	0.00	0.00	14
2	0.00	0.00	0.00	2
3	0.28	1.00	0.44	27
4	0.00	0.00	0.00	2
5	0.00	0.00	0.00	20
6	0.00	0.00	0.00	6
7	0.00	0.00	0.00	4
8	0.00	0.00	0.00	4
9	0.00	0.00	0.00	12
accuracy			0.27	100
macro avg	0.03	0.10	0.04	100
weighted avg	0.08	0.27	0.12	100

Output 3

```
              precision    recall  f1-score   support

           2       0.00      0.00      0.00         8
           3       0.00      0.00      0.00         5
           4       0.00      0.00      0.00         5
           5       0.00      0.00      0.00         3
           6       0.00      0.00      0.00         1
           7       0.34      1.00      0.50        32
           8       0.00      0.00      0.00         2
           9       0.00      0.00      0.00         7
          11       0.00      0.00      0.00        14
          12       0.20      0.10      0.13        10
          13       0.00      0.00      0.00         4
          14       0.00      0.00      0.00         9

    accuracy                           0.33       100
   macro avg       0.04      0.09      0.05       100
weighted avg       0.13      0.33      0.17       100

/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565:
UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels
with no predicted samples. Use `zero_division` parameter to control this
behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565:
UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels
with no predicted samples. Use `zero_division` parameter to control this
behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565:
UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels
with no predicted samples. Use `zero_division` parameter to control this
behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565:
UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels
with no predicted samples. Use `zero_division` parameter to control this
behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565:
UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels
with no predicted samples. Use `zero_division` parameter to control this
behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565:
UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels
```

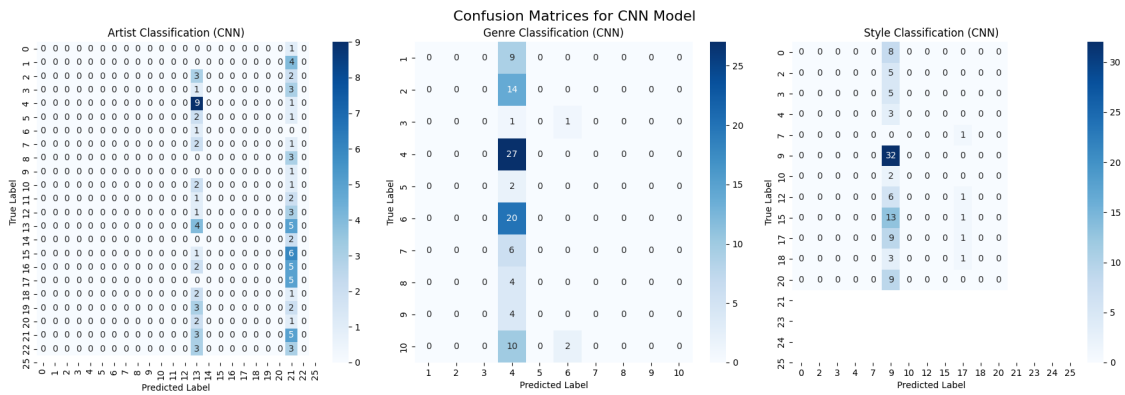
```
with no predicted samples. Use `zero_division` parameter to control this
behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565:
UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels
with no predicted samples. Use `zero_division` parameter to control this
behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565:
UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels
with no predicted samples. Use `zero_division` parameter to control this
behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565:
UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels
with no predicted samples. Use `zero_division` parameter to control this
behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

From here, we can infer that there is a severe class imbalance and the model predicts more dominant classes

2.3.4 Confusion Matrix for CNN

[ ]:
```
from sklearn.metrics import confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

all_class_names = {
    'artist': artist_classes,
    'genre': genre_classes,
    'style': style_classes
}

output_titles = ['Artist Classification (CNN)', 'Genre Classification (CNN)',␣
 ↪'Style Classification (CNN)']

plt.figure(figsize=(18, 6))

for i in range(3):
    cm_cnn = confusion_matrix(y_true_np[i], y_pred_classes[i])

    current_class_names = all_class_names[['artist', 'genre', 'style'][i]]

    plt.subplot(1, 3, i + 1)
    sns.heatmap(cm_cnn, annot=True, fmt='d', cmap='Blues',
                xticklabels=current_class_names,␣
 ↪yticklabels=current_class_names)
```

```
    plt.title(output_titles[i])
    plt.xlabel('Predicted Label')
    plt.ylabel('True Label')

plt.tight_layout()
plt.suptitle('Confusion Matrices for CNN Model', y=1.02, fontsize=16)
plt.show()
```



- metrics used : classification report, f1_score, class distributions, confusion matrix
- since the dataset size is too small (1000 images) which is not enough for transfer learning, we get a very low f1_score and a biased output for a few classes
- f1_score emerged out to be the best evaluation metric here because accuracy was giving better results (which is not true and we can't rely on that only)
- to fix this issues slightly, i tried lowering the learning rate of optimization function and the f1_score rised by 0.01
- to address the class imbalance issue, assigning higher weights to minority classes in the loss fucntion can be done
- data augmentation can also be done to increase the size of dataset

2.3.5 Outlier detection for CNN

[ ]:
```
import numpy as np
import matplotlib.pyplot as plt

all_images = []
all_labels = []
for images_batch, labels_batch in test_data_tf:
    all_images.append(images_batch.numpy())
    all_labels.append(labels_batch[0].numpy())
all_images = np.concatenate(all_images, axis=0)
all_labels = np.concatenate(all_labels, axis=0)
```

```
artist_preds_probs = model.predict(test_data_tf, verbose=0)[0]


misclassified_indices_cnn = np.where(y_true_np[0] != y_pred_classes[0])[0]

print(f"Total misclassified samples for Artist (CNN):␣
↪{len(misclassified_indices_cnn)}")

num_to_display = min(5, len(misclassified_indices_cnn))

if num_to_display > 0:
    np.random.shuffle(misclassified_indices_cnn)
    display_indices = misclassified_indices_cnn[:num_to_display]

    plt.figure(figsize=(15, 5))
    for i, idx in enumerate(display_indices):
        true_label = y_true_np[0][idx]
        predicted_label = y_pred_classes[0][idx]
        confidence = artist_preds_probs[idx, predicted_label]

        plt.subplot(1, num_to_display, i + 1)
        plt.imshow(all_images[idx])
        plt.axis('off')
        plt.title(f"True: {true_label}\nPred: {predicted_label}\nConf:␣
↪{confidence:.2f}", fontsize=10)
    plt.tight_layout()
    plt.suptitle('CNN Random Misclassifications (Artist)', y=1.05, fontsize=16)
    plt.show()
else:
    print("No misclassified samples found to display for CNN.")
```

```
Total misclassified samples for Artist (CNN): 91
```

The above images show where the model made an incorrect classification for the given artist.

The displayed confidence reflects the probability assigned to the model's (incorrect) predicted class.

2.3.6 Analysis of CNN Model :

After training, a deeper analysis using F1-scores, class distribution, and classification report was performed

Conclusion for CNN: The CNN model, despite using transfer learning, struggled significantly due to the small dataset size and severe class imbalance, leading to a biased and poor performance in feature learning for diverse categories.

3 Now, moving towards defining the CRNN architecture

3.1 CRNN Model : Combining CNN with Recurrent Layers

I continued to use the ResNet50 CNN backbone from above for the CRNN architecture for feature extraction.

While finding the optimum RNN model for this task, i found out that LSTM and GRU would be the best.

Moreover, GRU trains faster and is more effective on small datasets (as such in this case)

So, I experimented with both models - CRNN using LSTM - CRNN using GRU

The width dimension of the CNN was treated as timesteps and the height and channel dimensions were flattened into features. Now, this 3D vector (converted from original 4D from the CNN) is now fed into the RNN layers and further 3 classification heads are used for classifying artist, genre and style.

The CRNN models are trained on 10 epochs with the same hyperparameters as the CNN model.

3.1.1 CRNN model using LSTM layers

[ ]:
```
## LSTM
import tensorflow as tf
from tensorflow.keras.layers import Input, Reshape, LSTM, Dense
from tensorflow.keras.models import Model
from tensorflow.keras.applications import ResNet50

resnet_base = ResNet50(weights='imagenet', include_top=False,␣
 ↪input_shape=IMAGE_SIZE + [3])

for layer in resnet_base.layers:
    layer.trainable = False

cnn_output = resnet_base.output

reshaped_features = Reshape((cnn_output.shape[1], cnn_output.shape[2] *␣
 ↪cnn_output.shape[3]))(cnn_output)
```

```
lstm_output = LSTM(256, name='lstm_features')(reshaped_features)

artist_output_crnn = Dense(artist_classes_num, activation='softmax',␣
↪name='artist_output_crnn')(lstm_output)
genre_output_crnn = Dense(genre_classes_num, activation='softmax',␣
↪name='genre_output_crnn')(lstm_output)
style_output_crnn = Dense(style_classes_num, activation='softmax',␣
↪name='style_output_crnn')(lstm_output)

crnn_model = Model(inputs=resnet_base.input, outputs=[artist_output_crnn,␣
↪genre_output_crnn, style_output_crnn])

print(crnn_model.summary())
```

Model: "functional_1"

Layer (type)	Output Shape	Param #	Connected to
input_layer_1 (InputLayer)	(None, 224, 224, 3)	0	-
conv1_pad (ZeroPadding2D)	(None, 230, 230, 3)	0	input_layer_1[0]…
conv1_conv (Conv2D)	(None, 112, 112, 64)	9,472	conv1_pad[0][0]
conv1_bn (BatchNormalizatio…	(None, 112, 112, 64)	256	conv1_conv[0][0]
conv1_relu (Activation)	(None, 112, 112, 64)	0	conv1_bn[0][0]
pool1_pad (ZeroPadding2D)	(None, 114, 114, 64)	0	conv1_relu[0][0]
pool1_pool (MaxPooling2D)	(None, 56, 56, 64)	0	pool1_pad[0][0]
conv2_block1_1_conv (Conv2D)	(None, 56, 56, 64)	4,160	pool1_pool[0][0]
conv2_block1_1_bn (BatchNormalizatio…	(None, 56, 56, 64)	256	conv2_block1_1_c…
conv2_block1_1_relu	(None, 56, 56,	0	conv2_block1_1_b…

```
(Activation)          64)

conv2_block1_2_conv   (None, 56, 56,          36,928   conv2_block1_1_r…
(Conv2D)              64)

conv2_block1_2_bn     (None, 56, 56,             256   conv2_block1_2_c…
(BatchNormalizatio…   64)

conv2_block1_2_relu   (None, 56, 56,               0   conv2_block1_2_b…
(Activation)          64)

conv2_block1_0_conv   (None, 56, 56,          16,640   pool1_pool[0][0]
(Conv2D)              256)

conv2_block1_3_conv   (None, 56, 56,          16,640   conv2_block1_2_r…
(Conv2D)              256)

conv2_block1_0_bn     (None, 56, 56,           1,024   conv2_block1_0_c…
(BatchNormalizatio…   256)

conv2_block1_3_bn     (None, 56, 56,           1,024   conv2_block1_3_c…
(BatchNormalizatio…   256)

conv2_block1_add      (None, 56, 56,               0   conv2_block1_0_b…
(Add)                 256)                             conv2_block1_3_b…

conv2_block1_out      (None, 56, 56,               0   conv2_block1_add…
(Activation)          256)

conv2_block2_1_conv   (None, 56, 56,          16,448   conv2_block1_out…
(Conv2D)              64)

conv2_block2_1_bn     (None, 56, 56,             256   conv2_block2_1_c…
(BatchNormalizatio…   64)

conv2_block2_1_relu   (None, 56, 56,               0   conv2_block2_1_b…
(Activation)          64)

conv2_block2_2_conv   (None, 56, 56,          36,928   conv2_block2_1_r…
(Conv2D)              64)

conv2_block2_2_bn     (None, 56, 56,             256   conv2_block2_2_c…
(BatchNormalizatio…   64)

conv2_block2_2_relu   (None, 56, 56,               0   conv2_block2_2_b…
(Activation)          64)

conv2_block2_3_conv   (None, 56, 56,          16,640   conv2_block2_2_r…
```

```
(Conv2D)                256)

conv2_block2_3_bn       (None, 56, 56,          1,024   conv2_block2_3_c…
(BatchNormalizatio…     256)

conv2_block2_add        (None, 56, 56,              0   conv2_block1_out…
(Add)                   256)                            conv2_block2_3_b…

conv2_block2_out        (None, 56, 56,              0   conv2_block2_add…
(Activation)            256)

conv2_block3_1_conv     (None, 56, 56,         16,448   conv2_block2_out…
(Conv2D)                64)

conv2_block3_1_bn       (None, 56, 56,            256   conv2_block3_1_c…
(BatchNormalizatio…     64)

conv2_block3_1_relu     (None, 56, 56,              0   conv2_block3_1_b…
(Activation)            64)

conv2_block3_2_conv     (None, 56, 56,         36,928   conv2_block3_1_r…
(Conv2D)                64)

conv2_block3_2_bn       (None, 56, 56,            256   conv2_block3_2_c…
(BatchNormalizatio…     64)

conv2_block3_2_relu     (None, 56, 56,              0   conv2_block3_2_b…
(Activation)            64)

conv2_block3_3_conv     (None, 56, 56,         16,640   conv2_block3_2_r…
(Conv2D)                256)

conv2_block3_3_bn       (None, 56, 56,          1,024   conv2_block3_3_c…
(BatchNormalizatio…     256)

conv2_block3_add        (None, 56, 56,              0   conv2_block2_out…
(Add)                   256)                            conv2_block3_3_b…

conv2_block3_out        (None, 56, 56,              0   conv2_block3_add…
(Activation)            256)

conv3_block1_1_conv     (None, 28, 28,         32,896   conv2_block3_out…
(Conv2D)                128)

conv3_block1_1_bn       (None, 28, 28,            512   conv3_block1_1_c…
(BatchNormalizatio…     128)

conv3_block1_1_relu     (None, 28, 28,              0   conv3_block1_1_b…
```

```
(Activation)          128)

conv3_block1_2_conv   (None, 28, 28,        147,584   conv3_block1_1_r…
(Conv2D)              128)

conv3_block1_2_bn     (None, 28, 28,            512   conv3_block1_2_c…
(BatchNormalizatio…   128)

conv3_block1_2_relu   (None, 28, 28,              0   conv3_block1_2_b…
(Activation)          128)

conv3_block1_0_conv   (None, 28, 28,        131,584   conv2_block3_out…
(Conv2D)              512)

conv3_block1_3_conv   (None, 28, 28,         66,048   conv3_block1_2_r…
(Conv2D)              512)

conv3_block1_0_bn     (None, 28, 28,          2,048   conv3_block1_0_c…
(BatchNormalizatio…   512)

conv3_block1_3_bn     (None, 28, 28,          2,048   conv3_block1_3_c…
(BatchNormalizatio…   512)

conv3_block1_add      (None, 28, 28,              0   conv3_block1_0_b…
(Add)                 512)                            conv3_block1_3_b…

conv3_block1_out      (None, 28, 28,              0   conv3_block1_add…
(Activation)          512)

conv3_block2_1_conv   (None, 28, 28,         65,664   conv3_block1_out…
(Conv2D)              128)

conv3_block2_1_bn     (None, 28, 28,            512   conv3_block2_1_c…
(BatchNormalizatio…   128)

conv3_block2_1_relu   (None, 28, 28,              0   conv3_block2_1_b…
(Activation)          128)

conv3_block2_2_conv   (None, 28, 28,        147,584   conv3_block2_1_r…
(Conv2D)              128)

conv3_block2_2_bn     (None, 28, 28,            512   conv3_block2_2_c…
(BatchNormalizatio…   128)

conv3_block2_2_relu   (None, 28, 28,              0   conv3_block2_2_b…
(Activation)          128)

conv3_block2_3_conv   (None, 28, 28,         66,048   conv3_block2_2_r…
```

```
(Conv2D)                512)

conv3_block2_3_bn       (None, 28, 28,            2,048   conv3_block2_3_c…
(BatchNormalizatio…     512)

conv3_block2_add        (None, 28, 28,                0   conv3_block1_out…
(Add)                   512)                              conv3_block2_3_b…

conv3_block2_out        (None, 28, 28,                0   conv3_block2_add…
(Activation)            512)

conv3_block3_1_conv     (None, 28, 28,           65,664   conv3_block2_out…
(Conv2D)                128)

conv3_block3_1_bn       (None, 28, 28,              512   conv3_block3_1_c…
(BatchNormalizatio…     128)

conv3_block3_1_relu     (None, 28, 28,                0   conv3_block3_1_b…
(Activation)            128)

conv3_block3_2_conv     (None, 28, 28,          147,584   conv3_block3_1_r…
(Conv2D)                128)

conv3_block3_2_bn       (None, 28, 28,              512   conv3_block3_2_c…
(BatchNormalizatio…     128)

conv3_block3_2_relu     (None, 28, 28,                0   conv3_block3_2_b…
(Activation)            128)

conv3_block3_3_conv     (None, 28, 28,           66,048   conv3_block3_2_r…
(Conv2D)                512)

conv3_block3_3_bn       (None, 28, 28,            2,048   conv3_block3_3_c…
(BatchNormalizatio…     512)

conv3_block3_add        (None, 28, 28,                0   conv3_block2_out…
(Add)                   512)                              conv3_block3_3_b…

conv3_block3_out        (None, 28, 28,                0   conv3_block3_add…
(Activation)            512)

conv3_block4_1_conv     (None, 28, 28,           65,664   conv3_block3_out…
(Conv2D)                128)

conv3_block4_1_bn       (None, 28, 28,              512   conv3_block4_1_c…
(BatchNormalizatio…     128)

conv3_block4_1_relu     (None, 28, 28,                0   conv3_block4_1_b…
```

(Activation)	128)		
conv3_block4_2_conv (Conv2D)	(None, 28, 28, 128)	147,584	conv3_block4_1_r…
conv3_block4_2_bn (BatchNormalizatio…	(None, 28, 28, 128)	512	conv3_block4_2_c…
conv3_block4_2_relu (Activation)	(None, 28, 28, 128)	0	conv3_block4_2_b…
conv3_block4_3_conv (Conv2D)	(None, 28, 28, 512)	66,048	conv3_block4_2_r…
conv3_block4_3_bn (BatchNormalizatio…	(None, 28, 28, 512)	2,048	conv3_block4_3_c…
conv3_block4_add (Add)	(None, 28, 28, 512)	0	conv3_block3_out… conv3_block4_3_b…
conv3_block4_out (Activation)	(None, 28, 28, 512)	0	conv3_block4_add…
conv4_block1_1_conv (Conv2D)	(None, 14, 14, 256)	131,328	conv3_block4_out…
conv4_block1_1_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block1_1_c…
conv4_block1_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block1_1_b…
conv4_block1_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block1_1_r…
conv4_block1_2_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block1_2_c…
conv4_block1_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block1_2_b…
conv4_block1_0_conv (Conv2D)	(None, 14, 14, 1024)	525,312	conv3_block4_out…
conv4_block1_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block1_2_r…
conv4_block1_0_bn	(None, 14, 14,	4,096	conv4_block1_0_c…

```
(BatchNormalizatio…  1024)

conv4_block1_3_bn    (None, 14, 14,          4,096  conv4_block1_3_c…
(BatchNormalizatio…  1024)

conv4_block1_add     (None, 14, 14,              0  conv4_block1_0_b…
(Add)                1024)                          conv4_block1_3_b…

conv4_block1_out     (None, 14, 14,              0  conv4_block1_add…
(Activation)         1024)

conv4_block2_1_conv  (None, 14, 14,        262,400  conv4_block1_out…
(Conv2D)             256)

conv4_block2_1_bn    (None, 14, 14,          1,024  conv4_block2_1_c…
(BatchNormalizatio…  256)

conv4_block2_1_relu  (None, 14, 14,              0  conv4_block2_1_b…
(Activation)         256)

conv4_block2_2_conv  (None, 14, 14,        590,080  conv4_block2_1_r…
(Conv2D)             256)

conv4_block2_2_bn    (None, 14, 14,          1,024  conv4_block2_2_c…
(BatchNormalizatio…  256)

conv4_block2_2_relu  (None, 14, 14,              0  conv4_block2_2_b…
(Activation)         256)

conv4_block2_3_conv  (None, 14, 14,        263,168  conv4_block2_2_r…
(Conv2D)             1024)

conv4_block2_3_bn    (None, 14, 14,          4,096  conv4_block2_3_c…
(BatchNormalizatio…  1024)

conv4_block2_add     (None, 14, 14,              0  conv4_block1_out…
(Add)                1024)                          conv4_block2_3_b…

conv4_block2_out     (None, 14, 14,              0  conv4_block2_add…
(Activation)         1024)

conv4_block3_1_conv  (None, 14, 14,        262,400  conv4_block2_out…
(Conv2D)             256)

conv4_block3_1_bn    (None, 14, 14,          1,024  conv4_block3_1_c…
(BatchNormalizatio…  256)

conv4_block3_1_relu  (None, 14, 14,              0  conv4_block3_1_b…
```

(Activation)	256)		
conv4_block3_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block3_1_r…
conv4_block3_2_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block3_2_c…
conv4_block3_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block3_2_b…
conv4_block3_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block3_2_r…
conv4_block3_3_bn (BatchNormalizatio…	(None, 14, 14, 1024)	4,096	conv4_block3_3_c…
conv4_block3_add (Add)	(None, 14, 14, 1024)	0	conv4_block2_out… conv4_block3_3_b…
conv4_block3_out (Activation)	(None, 14, 14, 1024)	0	conv4_block3_add…
conv4_block4_1_conv (Conv2D)	(None, 14, 14, 256)	262,400	conv4_block3_out…
conv4_block4_1_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block4_1_c…
conv4_block4_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block4_1_b…
conv4_block4_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block4_1_r…
conv4_block4_2_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block4_2_c…
conv4_block4_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block4_2_b…
conv4_block4_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block4_2_r…
conv4_block4_3_bn (BatchNormalizatio…	(None, 14, 14, 1024)	4,096	conv4_block4_3_c…
conv4_block4_add	(None, 14, 14,	0	conv4_block3_out…

(Add)	1024)		conv4_block4_3_b…
conv4_block4_out (Activation)	(None, 14, 14, 1024)	0	conv4_block4_add…
conv4_block5_1_conv (Conv2D)	(None, 14, 14, 256)	262,400	conv4_block4_out…
conv4_block5_1_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block5_1_c…
conv4_block5_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block5_1_b…
conv4_block5_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block5_1_r…
conv4_block5_2_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block5_2_c…
conv4_block5_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block5_2_b…
conv4_block5_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block5_2_r…
conv4_block5_3_bn (BatchNormalizatio…	(None, 14, 14, 1024)	4,096	conv4_block5_3_c…
conv4_block5_add (Add)	(None, 14, 14, 1024)	0	conv4_block4_out… conv4_block5_3_b…
conv4_block5_out (Activation)	(None, 14, 14, 1024)	0	conv4_block5_add…
conv4_block6_1_conv (Conv2D)	(None, 14, 14, 256)	262,400	conv4_block5_out…
conv4_block6_1_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block6_1_c…
conv4_block6_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block6_1_b…
conv4_block6_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block6_1_r…
conv4_block6_2_bn	(None, 14, 14,	1,024	conv4_block6_2_c…

(BatchNormalizatio…	256)		
conv4_block6_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block6_2_b…
conv4_block6_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block6_2_r…
conv4_block6_3_bn (BatchNormalizatio…	(None, 14, 14, 1024)	4,096	conv4_block6_3_c…
conv4_block6_add (Add)	(None, 14, 14, 1024)	0	conv4_block5_out… conv4_block6_3_b…
conv4_block6_out (Activation)	(None, 14, 14, 1024)	0	conv4_block6_add…
conv5_block1_1_conv (Conv2D)	(None, 7, 7, 512)	524,800	conv4_block6_out…
conv5_block1_1_bn (BatchNormalizatio…	(None, 7, 7, 512)	2,048	conv5_block1_1_c…
conv5_block1_1_relu (Activation)	(None, 7, 7, 512)	0	conv5_block1_1_b…
conv5_block1_2_conv (Conv2D)	(None, 7, 7, 512)	2,359,808	conv5_block1_1_r…
conv5_block1_2_bn (BatchNormalizatio…	(None, 7, 7, 512)	2,048	conv5_block1_2_c…
conv5_block1_2_relu (Activation)	(None, 7, 7, 512)	0	conv5_block1_2_b…
conv5_block1_0_conv (Conv2D)	(None, 7, 7, 2048)	2,099,200	conv4_block6_out…
conv5_block1_3_conv (Conv2D)	(None, 7, 7, 2048)	1,050,624	conv5_block1_2_r…
conv5_block1_0_bn (BatchNormalizatio…	(None, 7, 7, 2048)	8,192	conv5_block1_0_c…
conv5_block1_3_bn (BatchNormalizatio…	(None, 7, 7, 2048)	8,192	conv5_block1_3_c…
conv5_block1_add	(None, 7, 7,	0	conv5_block1_0_b…

(Add)	2048)		conv5_block1_3_b…
conv5_block1_out (Activation)	(None, 7, 7, 2048)	0	conv5_block1_add…
conv5_block2_1_conv (Conv2D)	(None, 7, 7, 512)	1,049,088	conv5_block1_out…
conv5_block2_1_bn (BatchNormalizatio…	(None, 7, 7, 512)	2,048	conv5_block2_1_c…
conv5_block2_1_relu (Activation)	(None, 7, 7, 512)	0	conv5_block2_1_b…
conv5_block2_2_conv (Conv2D)	(None, 7, 7, 512)	2,359,808	conv5_block2_1_r…
conv5_block2_2_bn (BatchNormalizatio…	(None, 7, 7, 512)	2,048	conv5_block2_2_c…
conv5_block2_2_relu (Activation)	(None, 7, 7, 512)	0	conv5_block2_2_b…
conv5_block2_3_conv (Conv2D)	(None, 7, 7, 2048)	1,050,624	conv5_block2_2_r…
conv5_block2_3_bn (BatchNormalizatio…	(None, 7, 7, 2048)	8,192	conv5_block2_3_c…
conv5_block2_add (Add)	(None, 7, 7, 2048)	0	conv5_block1_out… conv5_block2_3_b…
conv5_block2_out (Activation)	(None, 7, 7, 2048)	0	conv5_block2_add…
conv5_block3_1_conv (Conv2D)	(None, 7, 7, 512)	1,049,088	conv5_block2_out…
conv5_block3_1_bn (BatchNormalizatio…	(None, 7, 7, 512)	2,048	conv5_block3_1_c…
conv5_block3_1_relu (Activation)	(None, 7, 7, 512)	0	conv5_block3_1_b…
conv5_block3_2_conv (Conv2D)	(None, 7, 7, 512)	2,359,808	conv5_block3_1_r…
conv5_block3_2_bn	(None, 7, 7, 512)	2,048	conv5_block3_2_c…

```
(BatchNormalizatio…

conv5_block3_2_relu   (None, 7, 7, 512)            0   conv5_block3_2_b…
(Activation)

conv5_block3_3_conv   (None, 7, 7,         1,050,624   conv5_block3_2_r…
(Conv2D)              2048)

conv5_block3_3_bn     (None, 7, 7,             8,192   conv5_block3_3_c…
(BatchNormalizatio…   2048)

conv5_block3_add      (None, 7, 7,                 0   conv5_block2_out…
(Add)                 2048)                            conv5_block3_3_b…

conv5_block3_out      (None, 7, 7,                 0   conv5_block3_add…
(Activation)          2048)

reshape (Reshape)     (None, 7, 14336)             0   conv5_block3_out…

lstm_features         (None, 256)         14,943,232   reshape[0][0]
(LSTM)

artist_output_crnn    (None, 24)               6,168   lstm_features[0]…
(Dense)

genre_output_crnn     (None, 10)               2,570   lstm_features[0]…
(Dense)

style_output_crnn     (None, 16)               4,112   lstm_features[0]…
(Dense)


Total params: 38,543,794 (147.03 MB)


Trainable params: 14,956,082 (57.05 MB)


Non-trainable params: 23,587,712 (89.98 MB)
```

None

[ ]:
```python
from tensorflow.keras.metrics import F1Score
# Compile the multi-output model

metrics ={
    "artist_output_crnn": ["accuracy"],
```

```
    "genre_output_crnn": ["accuracy"],
    "style_output_crnn": ["accuracy"]
}

optimizer = tf.keras.optimizers.Adam(learning_rate=1e-4)

crnn_model.compile(
    optimizer=optimizer,
    loss={
        'artist_output_crnn': 'sparse_categorical_crossentropy',
        'genre_output_crnn': 'sparse_categorical_crossentropy',
        'style_output_crnn': 'sparse_categorical_crossentropy'
    },
    metrics = metrics
)

print("Model compiled successfully!")
```

```
Model compiled successfully!
```

[ ]:
```
history_crnn = crnn_model.fit(
    train_data_tf,
    validation_data=val_data_tf,
    epochs=10
)
```

```
Epoch 1/10
25/25              31s 459ms/step -
artist_output_crnn_accuracy: 0.0915 - artist_output_crnn_loss: 3.1271 -
genre_output_crnn_accuracy: 0.1814 - genre_output_crnn_loss: 2.2356 - loss:
7.7567 - style_output_crnn_accuracy: 0.2318 - style_output_crnn_loss: 2.3940 -
val_artist_output_crnn_accuracy: 0.0700 - val_artist_output_crnn_loss: 3.1271 -
val_genre_output_crnn_accuracy: 0.1700 - val_genre_output_crnn_loss: 2.1801 -
val_loss: 7.3382 - val_style_output_crnn_accuracy: 0.3800 -
val_style_output_crnn_loss: 2.1017
Epoch 2/10
25/25              6s 241ms/step -
artist_output_crnn_accuracy: 0.1054 - artist_output_crnn_loss: 3.0053 -
genre_output_crnn_accuracy: 0.2129 - genre_output_crnn_loss: 2.1404 - loss:
7.2966 - style_output_crnn_accuracy: 0.3234 - style_output_crnn_loss: 2.1509 -
val_artist_output_crnn_accuracy: 0.0700 - val_artist_output_crnn_loss: 3.1183 -
val_genre_output_crnn_accuracy: 0.1600 - val_genre_output_crnn_loss: 2.1677 -
val_loss: 7.2889 - val_style_output_crnn_accuracy: 0.3900 -
val_style_output_crnn_loss: 2.0580
Epoch 3/10
25/25              7s 268ms/step -
artist_output_crnn_accuracy: 0.1079 - artist_output_crnn_loss: 3.0265 -
genre_output_crnn_accuracy: 0.2614 - genre_output_crnn_loss: 2.0823 - loss:
```

```
7.2293 - style_output_crnn_accuracy: 0.3190 - style_output_crnn_loss: 2.1205 -
val_artist_output_crnn_accuracy: 0.0900 - val_artist_output_crnn_loss: 3.1046 -
val_genre_output_crnn_accuracy: 0.2000 - val_genre_output_crnn_loss: 2.1202 -
val_loss: 7.2016 - val_style_output_crnn_accuracy: 0.3300 -
val_style_output_crnn_loss: 2.0167
Epoch 4/10
25/25              6s 243ms/step -
artist_output_crnn_accuracy: 0.1149 - artist_output_crnn_loss: 2.9720 -
genre_output_crnn_accuracy: 0.2419 - genre_output_crnn_loss: 2.0898 - loss:
7.1566 - style_output_crnn_accuracy: 0.2888 - style_output_crnn_loss: 2.0948 -
val_artist_output_crnn_accuracy: 0.0900 - val_artist_output_crnn_loss: 3.0820 -
val_genre_output_crnn_accuracy: 0.2100 - val_genre_output_crnn_loss: 2.0956 -
val_loss: 7.1309 - val_style_output_crnn_accuracy: 0.3900 -
val_style_output_crnn_loss: 1.9933
Epoch 5/10
25/25              7s 261ms/step -
artist_output_crnn_accuracy: 0.1454 - artist_output_crnn_loss: 2.9247 -
genre_output_crnn_accuracy: 0.2613 - genre_output_crnn_loss: 2.0793 - loss:
7.0948 - style_output_crnn_accuracy: 0.3161 - style_output_crnn_loss: 2.0907 -
val_artist_output_crnn_accuracy: 0.1600 - val_artist_output_crnn_loss: 3.0325 -
val_genre_output_crnn_accuracy: 0.1900 - val_genre_output_crnn_loss: 2.1127 -
val_loss: 7.0043 - val_style_output_crnn_accuracy: 0.4000 -
val_style_output_crnn_loss: 1.9389
Epoch 6/10
25/25              6s 237ms/step -
artist_output_crnn_accuracy: 0.1765 - artist_output_crnn_loss: 2.8864 -
genre_output_crnn_accuracy: 0.2508 - genre_output_crnn_loss: 2.0535 - loss:
6.9854 - style_output_crnn_accuracy: 0.3217 - style_output_crnn_loss: 2.0456 -
val_artist_output_crnn_accuracy: 0.1600 - val_artist_output_crnn_loss: 3.0490 -
val_genre_output_crnn_accuracy: 0.2200 - val_genre_output_crnn_loss: 2.0278 -
val_loss: 6.9457 - val_style_output_crnn_accuracy: 0.3800 -
val_style_output_crnn_loss: 1.8899
Epoch 7/10
25/25              7s 262ms/step -
artist_output_crnn_accuracy: 0.1472 - artist_output_crnn_loss: 2.8676 -
genre_output_crnn_accuracy: 0.2897 - genre_output_crnn_loss: 2.0048 - loss:
6.8942 - style_output_crnn_accuracy: 0.3598 - style_output_crnn_loss: 2.0218 -
val_artist_output_crnn_accuracy: 0.1000 - val_artist_output_crnn_loss: 3.0511 -
val_genre_output_crnn_accuracy: 0.2200 - val_genre_output_crnn_loss: 2.1269 -
val_loss: 6.9602 - val_style_output_crnn_accuracy: 0.4000 -
val_style_output_crnn_loss: 1.8269
Epoch 8/10
25/25              6s 240ms/step -
artist_output_crnn_accuracy: 0.1783 - artist_output_crnn_loss: 2.8235 -
genre_output_crnn_accuracy: 0.2790 - genre_output_crnn_loss: 2.0076 - loss:
6.7651 - style_output_crnn_accuracy: 0.3622 - style_output_crnn_loss: 1.9341 -
val_artist_output_crnn_accuracy: 0.1600 - val_artist_output_crnn_loss: 2.9230 -
val_genre_output_crnn_accuracy: 0.2600 - val_genre_output_crnn_loss: 2.0327 -
```

```
val_loss: 6.8433 - val_style_output_crnn_accuracy: 0.4200 -
val_style_output_crnn_loss: 1.8096
Epoch 9/10
25/25              7s 262ms/step -
artist_output_crnn_accuracy: 0.1897 - artist_output_crnn_loss: 2.7831 -
genre_output_crnn_accuracy: 0.3010 - genre_output_crnn_loss: 1.9835 - loss:
6.7305 - style_output_crnn_accuracy: 0.3760 - style_output_crnn_loss: 1.9639 -
val_artist_output_crnn_accuracy: 0.1600 - val_artist_output_crnn_loss: 2.9510 -
val_genre_output_crnn_accuracy: 0.3000 - val_genre_output_crnn_loss: 1.9679 -
val_loss: 6.9001 - val_style_output_crnn_accuracy: 0.3800 -
val_style_output_crnn_loss: 1.9263
Epoch 10/10
25/25              6s 236ms/step -
artist_output_crnn_accuracy: 0.2185 - artist_output_crnn_loss: 2.7901 -
genre_output_crnn_accuracy: 0.3332 - genre_output_crnn_loss: 1.9510 - loss:
6.7159 - style_output_crnn_accuracy: 0.3402 - style_output_crnn_loss: 1.9748 -
val_artist_output_crnn_accuracy: 0.1400 - val_artist_output_crnn_loss: 2.9544 -
val_genre_output_crnn_accuracy: 0.2800 - val_genre_output_crnn_loss: 2.0320 -
val_loss: 6.7861 - val_style_output_crnn_accuracy: 0.4100 -
val_style_output_crnn_loss: 1.8548
```

3.2 CRNN Model using GRU Layers

CRNN model using Gated Recurrent Units (GRU) instead of Long Short-Term Memory (LSTM) units. The CNN backbone (ResNet50) remains the same for feature extraction.

[ ]:
```
import tensorflow as tf
from tensorflow.keras.layers import Input, Reshape, GRU, Dense
from tensorflow.keras.models import Model
from tensorflow.keras.applications import ResNet50

resnet_base_gru = ResNet50(weights='imagenet', include_top=False,␣
↪input_shape=IMAGE_SIZE + [3])

for layer in resnet_base_gru.layers:
    layer.trainable = False

cnn_output_gru = resnet_base_gru.output

reshaped_features_gru = Reshape((cnn_output_gru.shape[1], cnn_output_gru.
↪shape[2] * cnn_output_gru.shape[3]))(cnn_output_gru)

gru_output = GRU(256, name='gru_features')(reshaped_features_gru)

artist_output_gru = Dense(artist_classes_num, activation='softmax',␣
↪name='artist_output_gru')(gru_output)
genre_output_gru = Dense(genre_classes_num, activation='softmax',␣
↪name='genre_output_gru')(gru_output)
```

```
style_output_gru = Dense(style_classes_num, activation='softmax',␣
↪name='style_output_gru')(gru_output)

gru_crnn_model = Model(inputs=resnet_base_gru.input,␣
↪outputs=[artist_output_gru, genre_output_gru, style_output_gru])

print(gru_crnn_model.summary())
```

Model: "functional_3"

Layer (type)	Output Shape	Param #	Connected to
input_layer_3 (InputLayer)	(None, 224, 224, 3)	0	-
conv1_pad (ZeroPadding2D)	(None, 230, 230, 3)	0	input_layer_3[0]…
conv1_conv (Conv2D)	(None, 112, 112, 64)	9,472	conv1_pad[0][0]
conv1_bn (BatchNormalizatio…	(None, 112, 112, 64)	256	conv1_conv[0][0]
conv1_relu (Activation)	(None, 112, 112, 64)	0	conv1_bn[0][0]
pool1_pad (ZeroPadding2D)	(None, 114, 114, 64)	0	conv1_relu[0][0]
pool1_pool (MaxPooling2D)	(None, 56, 56, 64)	0	pool1_pad[0][0]
conv2_block1_1_conv (Conv2D)	(None, 56, 56, 64)	4,160	pool1_pool[0][0]
conv2_block1_1_bn (BatchNormalizatio…	(None, 56, 56, 64)	256	conv2_block1_1_c…
conv2_block1_1_relu (Activation)	(None, 56, 56, 64)	0	conv2_block1_1_b…
conv2_block1_2_conv (Conv2D)	(None, 56, 56, 64)	36,928	conv2_block1_1_r…
conv2_block1_2_bn	(None, 56, 56,	256	conv2_block1_2_c…

```
(BatchNormalizatio…  64)

conv2_block1_2_relu  (None, 56, 56,              0  conv2_block1_2_b…
(Activation)         64)

conv2_block1_0_conv  (None, 56, 56,         16,640  pool1_pool[0][0]
(Conv2D)             256)

conv2_block1_3_conv  (None, 56, 56,         16,640  conv2_block1_2_r…
(Conv2D)             256)

conv2_block1_0_bn    (None, 56, 56,          1,024  conv2_block1_0_c…
(BatchNormalizatio…  256)

conv2_block1_3_bn    (None, 56, 56,          1,024  conv2_block1_3_c…
(BatchNormalizatio…  256)

conv2_block1_add     (None, 56, 56,              0  conv2_block1_0_b…
(Add)                256)                           conv2_block1_3_b…

conv2_block1_out     (None, 56, 56,              0  conv2_block1_add…
(Activation)         256)

conv2_block2_1_conv  (None, 56, 56,         16,448  conv2_block1_out…
(Conv2D)             64)

conv2_block2_1_bn    (None, 56, 56,            256  conv2_block2_1_c…
(BatchNormalizatio…  64)

conv2_block2_1_relu  (None, 56, 56,              0  conv2_block2_1_b…
(Activation)         64)

conv2_block2_2_conv  (None, 56, 56,         36,928  conv2_block2_1_r…
(Conv2D)             64)

conv2_block2_2_bn    (None, 56, 56,            256  conv2_block2_2_c…
(BatchNormalizatio…  64)

conv2_block2_2_relu  (None, 56, 56,              0  conv2_block2_2_b…
(Activation)         64)

conv2_block2_3_conv  (None, 56, 56,         16,640  conv2_block2_2_r…
(Conv2D)             256)

conv2_block2_3_bn    (None, 56, 56,          1,024  conv2_block2_3_c…
(BatchNormalizatio…  256)

conv2_block2_add     (None, 56, 56,              0  conv2_block1_out…
```

(Add)	256)		conv2_block2_3_b…
conv2_block2_out (Activation)	(None, 56, 56, 256)	0	conv2_block2_add…
conv2_block3_1_conv (Conv2D)	(None, 56, 56, 64)	16,448	conv2_block2_out…
conv2_block3_1_bn (BatchNormalizatio…	(None, 56, 56, 64)	256	conv2_block3_1_c…
conv2_block3_1_relu (Activation)	(None, 56, 56, 64)	0	conv2_block3_1_b…
conv2_block3_2_conv (Conv2D)	(None, 56, 56, 64)	36,928	conv2_block3_1_r…
conv2_block3_2_bn (BatchNormalizatio…	(None, 56, 56, 64)	256	conv2_block3_2_c…
conv2_block3_2_relu (Activation)	(None, 56, 56, 64)	0	conv2_block3_2_b…
conv2_block3_3_conv (Conv2D)	(None, 56, 56, 256)	16,640	conv2_block3_2_r…
conv2_block3_3_bn (BatchNormalizatio…	(None, 56, 56, 256)	1,024	conv2_block3_3_c…
conv2_block3_add (Add)	(None, 56, 56, 256)	0	conv2_block2_out… conv2_block3_3_b…
conv2_block3_out (Activation)	(None, 56, 56, 256)	0	conv2_block3_add…
conv3_block1_1_conv (Conv2D)	(None, 28, 28, 128)	32,896	conv2_block3_out…
conv3_block1_1_bn (BatchNormalizatio…	(None, 28, 28, 128)	512	conv3_block1_1_c…
conv3_block1_1_relu (Activation)	(None, 28, 28, 128)	0	conv3_block1_1_b…
conv3_block1_2_conv (Conv2D)	(None, 28, 28, 128)	147,584	conv3_block1_1_r…
conv3_block1_2_bn	(None, 28, 28,	512	conv3_block1_2_c…

```
(BatchNormalizatio…  128)

conv3_block1_2_relu  (None, 28, 28,              0  conv3_block1_2_b…
(Activation)         128)

conv3_block1_0_conv  (None, 28, 28,        131,584  conv2_block3_out…
(Conv2D)             512)

conv3_block1_3_conv  (None, 28, 28,         66,048  conv3_block1_2_r…
(Conv2D)             512)

conv3_block1_0_bn    (None, 28, 28,          2,048  conv3_block1_0_c…
(BatchNormalizatio…  512)

conv3_block1_3_bn    (None, 28, 28,          2,048  conv3_block1_3_c…
(BatchNormalizatio…  512)

conv3_block1_add     (None, 28, 28,              0  conv3_block1_0_b…
(Add)                512)                           conv3_block1_3_b…

conv3_block1_out     (None, 28, 28,              0  conv3_block1_add…
(Activation)         512)

conv3_block2_1_conv  (None, 28, 28,         65,664  conv3_block1_out…
(Conv2D)             128)

conv3_block2_1_bn    (None, 28, 28,            512  conv3_block2_1_c…
(BatchNormalizatio…  128)

conv3_block2_1_relu  (None, 28, 28,              0  conv3_block2_1_b…
(Activation)         128)

conv3_block2_2_conv  (None, 28, 28,        147,584  conv3_block2_1_r…
(Conv2D)             128)

conv3_block2_2_bn    (None, 28, 28,            512  conv3_block2_2_c…
(BatchNormalizatio…  128)

conv3_block2_2_relu  (None, 28, 28,              0  conv3_block2_2_b…
(Activation)         128)

conv3_block2_3_conv  (None, 28, 28,         66,048  conv3_block2_2_r…
(Conv2D)             512)

conv3_block2_3_bn    (None, 28, 28,          2,048  conv3_block2_3_c…
(BatchNormalizatio…  512)

conv3_block2_add     (None, 28, 28,              0  conv3_block1_out…
```

(Add)	512)		conv3_block2_3_b…
conv3_block2_out (Activation)	(None, 28, 28, 512)	0	conv3_block2_add…
conv3_block3_1_conv (Conv2D)	(None, 28, 28, 128)	65,664	conv3_block2_out…
conv3_block3_1_bn (BatchNormalizatio…	(None, 28, 28, 128)	512	conv3_block3_1_c…
conv3_block3_1_relu (Activation)	(None, 28, 28, 128)	0	conv3_block3_1_b…
conv3_block3_2_conv (Conv2D)	(None, 28, 28, 128)	147,584	conv3_block3_1_r…
conv3_block3_2_bn (BatchNormalizatio…	(None, 28, 28, 128)	512	conv3_block3_2_c…
conv3_block3_2_relu (Activation)	(None, 28, 28, 128)	0	conv3_block3_2_b…
conv3_block3_3_conv (Conv2D)	(None, 28, 28, 512)	66,048	conv3_block3_2_r…
conv3_block3_3_bn (BatchNormalizatio…	(None, 28, 28, 512)	2,048	conv3_block3_3_c…
conv3_block3_add (Add)	(None, 28, 28, 512)	0	conv3_block2_out… conv3_block3_3_b…
conv3_block3_out (Activation)	(None, 28, 28, 512)	0	conv3_block3_add…
conv3_block4_1_conv (Conv2D)	(None, 28, 28, 128)	65,664	conv3_block3_out…
conv3_block4_1_bn (BatchNormalizatio…	(None, 28, 28, 128)	512	conv3_block4_1_c…
conv3_block4_1_relu (Activation)	(None, 28, 28, 128)	0	conv3_block4_1_b…
conv3_block4_2_conv (Conv2D)	(None, 28, 28, 128)	147,584	conv3_block4_1_r…
conv3_block4_2_bn	(None, 28, 28,	512	conv3_block4_2_c…

(BatchNormalizatio…	128)		
conv3_block4_2_relu (Activation)	(None, 28, 28, 128)	0	conv3_block4_2_b…
conv3_block4_3_conv (Conv2D)	(None, 28, 28, 512)	66,048	conv3_block4_2_r…
conv3_block4_3_bn (BatchNormalizatio…	(None, 28, 28, 512)	2,048	conv3_block4_3_c…
conv3_block4_add (Add)	(None, 28, 28, 512)	0	conv3_block3_out… conv3_block4_3_b…
conv3_block4_out (Activation)	(None, 28, 28, 512)	0	conv3_block4_add…
conv4_block1_1_conv (Conv2D)	(None, 14, 14, 256)	131,328	conv3_block4_out…
conv4_block1_1_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block1_1_c…
conv4_block1_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block1_1_b…
conv4_block1_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block1_1_r…
conv4_block1_2_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block1_2_c…
conv4_block1_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block1_2_b…
conv4_block1_0_conv (Conv2D)	(None, 14, 14, 1024)	525,312	conv3_block4_out…
conv4_block1_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block1_2_r…
conv4_block1_0_bn (BatchNormalizatio…	(None, 14, 14, 1024)	4,096	conv4_block1_0_c…
conv4_block1_3_bn (BatchNormalizatio…	(None, 14, 14, 1024)	4,096	conv4_block1_3_c…
conv4_block1_add	(None, 14, 14,	0	conv4_block1_0_b…

(Add)	1024)		conv4_block1_3_b…
conv4_block1_out (Activation)	(None, 14, 14, 1024)	0	conv4_block1_add…
conv4_block2_1_conv (Conv2D)	(None, 14, 14, 256)	262,400	conv4_block1_out…
conv4_block2_1_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block2_1_c…
conv4_block2_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block2_1_b…
conv4_block2_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block2_1_r…
conv4_block2_2_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block2_2_c…
conv4_block2_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block2_2_b…
conv4_block2_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block2_2_r…
conv4_block2_3_bn (BatchNormalizatio…	(None, 14, 14, 1024)	4,096	conv4_block2_3_c…
conv4_block2_add (Add)	(None, 14, 14, 1024)	0	conv4_block1_out… conv4_block2_3_b…
conv4_block2_out (Activation)	(None, 14, 14, 1024)	0	conv4_block2_add…
conv4_block3_1_conv (Conv2D)	(None, 14, 14, 256)	262,400	conv4_block2_out…
conv4_block3_1_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block3_1_c…
conv4_block3_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block3_1_b…
conv4_block3_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block3_1_r…
conv4_block3_2_bn	(None, 14, 14,	1,024	conv4_block3_2_c…

```
(BatchNormalizatio…   256)

conv4_block3_2_relu   (None, 14, 14,              0   conv4_block3_2_b…
(Activation)          256)

conv4_block3_3_conv   (None, 14, 14,        263,168   conv4_block3_2_r…
(Conv2D)              1024)

conv4_block3_3_bn     (None, 14, 14,          4,096   conv4_block3_3_c…
(BatchNormalizatio…   1024)

conv4_block3_add      (None, 14, 14,              0   conv4_block2_out…
(Add)                 1024)                           conv4_block3_3_b…

conv4_block3_out      (None, 14, 14,              0   conv4_block3_add…
(Activation)          1024)

conv4_block4_1_conv   (None, 14, 14,        262,400   conv4_block3_out…
(Conv2D)              256)

conv4_block4_1_bn     (None, 14, 14,          1,024   conv4_block4_1_c…
(BatchNormalizatio…   256)

conv4_block4_1_relu   (None, 14, 14,              0   conv4_block4_1_b…
(Activation)          256)

conv4_block4_2_conv   (None, 14, 14,        590,080   conv4_block4_1_r…
(Conv2D)              256)

conv4_block4_2_bn     (None, 14, 14,          1,024   conv4_block4_2_c…
(BatchNormalizatio…   256)

conv4_block4_2_relu   (None, 14, 14,              0   conv4_block4_2_b…
(Activation)          256)

conv4_block4_3_conv   (None, 14, 14,        263,168   conv4_block4_2_r…
(Conv2D)              1024)

conv4_block4_3_bn     (None, 14, 14,          4,096   conv4_block4_3_c…
(BatchNormalizatio…   1024)

conv4_block4_add      (None, 14, 14,              0   conv4_block3_out…
(Add)                 1024)                           conv4_block4_3_b…

conv4_block4_out      (None, 14, 14,              0   conv4_block4_add…
(Activation)          1024)

conv4_block5_1_conv   (None, 14, 14,        262,400   conv4_block4_out…
```

```
(Conv2D)                256)

conv4_block5_1_bn       (None, 14, 14,            1,024   conv4_block5_1_c…
(BatchNormalizatio…     256)

conv4_block5_1_relu     (None, 14, 14,                0   conv4_block5_1_b…
(Activation)            256)

conv4_block5_2_conv     (None, 14, 14,          590,080   conv4_block5_1_r…
(Conv2D)                256)

conv4_block5_2_bn       (None, 14, 14,            1,024   conv4_block5_2_c…
(BatchNormalizatio…     256)

conv4_block5_2_relu     (None, 14, 14,                0   conv4_block5_2_b…
(Activation)            256)

conv4_block5_3_conv     (None, 14, 14,          263,168   conv4_block5_2_r…
(Conv2D)                1024)

conv4_block5_3_bn       (None, 14, 14,            4,096   conv4_block5_3_c…
(BatchNormalizatio…     1024)

conv4_block5_add        (None, 14, 14,                0   conv4_block4_out…
(Add)                   1024)                             conv4_block5_3_b…

conv4_block5_out        (None, 14, 14,                0   conv4_block5_add…
(Activation)            1024)

conv4_block6_1_conv     (None, 14, 14,          262,400   conv4_block5_out…
(Conv2D)                256)

conv4_block6_1_bn       (None, 14, 14,            1,024   conv4_block6_1_c…
(BatchNormalizatio…     256)

conv4_block6_1_relu     (None, 14, 14,                0   conv4_block6_1_b…
(Activation)            256)

conv4_block6_2_conv     (None, 14, 14,          590,080   conv4_block6_1_r…
(Conv2D)                256)

conv4_block6_2_bn       (None, 14, 14,            1,024   conv4_block6_2_c…
(BatchNormalizatio…     256)

conv4_block6_2_relu     (None, 14, 14,                0   conv4_block6_2_b…
(Activation)            256)

conv4_block6_3_conv     (None, 14, 14,          263,168   conv4_block6_2_r…
```

```
(Conv2D)                1024)

conv4_block6_3_bn       (None, 14, 14,          4,096   conv4_block6_3_c…
(BatchNormalizatio…     1024)

conv4_block6_add        (None, 14, 14,              0   conv4_block5_out…
(Add)                   1024)                           conv4_block6_3_b…

conv4_block6_out        (None, 14, 14,              0   conv4_block6_add…
(Activation)            1024)

conv5_block1_1_conv     (None, 7, 7, 512)     524,800   conv4_block6_out…
(Conv2D)

conv5_block1_1_bn       (None, 7, 7, 512)       2,048   conv5_block1_1_c…
(BatchNormalizatio…

conv5_block1_1_relu     (None, 7, 7, 512)           0   conv5_block1_1_b…
(Activation)

conv5_block1_2_conv     (None, 7, 7, 512)   2,359,808   conv5_block1_1_r…
(Conv2D)

conv5_block1_2_bn       (None, 7, 7, 512)       2,048   conv5_block1_2_c…
(BatchNormalizatio…

conv5_block1_2_relu     (None, 7, 7, 512)           0   conv5_block1_2_b…
(Activation)

conv5_block1_0_conv     (None, 7, 7,        2,099,200   conv4_block6_out…
(Conv2D)                2048)

conv5_block1_3_conv     (None, 7, 7,        1,050,624   conv5_block1_2_r…
(Conv2D)                2048)

conv5_block1_0_bn       (None, 7, 7,            8,192   conv5_block1_0_c…
(BatchNormalizatio…     2048)

conv5_block1_3_bn       (None, 7, 7,            8,192   conv5_block1_3_c…
(BatchNormalizatio…     2048)

conv5_block1_add        (None, 7, 7,                0   conv5_block1_0_b…
(Add)                   2048)                           conv5_block1_3_b…

conv5_block1_out        (None, 7, 7,                0   conv5_block1_add…
(Activation)            2048)

conv5_block2_1_conv     (None, 7, 7, 512)   1,049,088   conv5_block1_out…
```

```
(Conv2D)

conv5_block2_1_bn     (None, 7, 7, 512)        2,048   conv5_block2_1_c…
(BatchNormalizatio…

conv5_block2_1_relu   (None, 7, 7, 512)            0   conv5_block2_1_b…
(Activation)

conv5_block2_2_conv   (None, 7, 7, 512)    2,359,808   conv5_block2_1_r…
(Conv2D)

conv5_block2_2_bn     (None, 7, 7, 512)        2,048   conv5_block2_2_c…
(BatchNormalizatio…

conv5_block2_2_relu   (None, 7, 7, 512)            0   conv5_block2_2_b…
(Activation)

conv5_block2_3_conv   (None, 7, 7,         1,050,624   conv5_block2_2_r…
(Conv2D)              2048)

conv5_block2_3_bn     (None, 7, 7,             8,192   conv5_block2_3_c…
(BatchNormalizatio…   2048)

conv5_block2_add      (None, 7, 7,                 0   conv5_block1_out…
(Add)                 2048)                            conv5_block2_3_b…

conv5_block2_out      (None, 7, 7,                 0   conv5_block2_add…
(Activation)          2048)

conv5_block3_1_conv   (None, 7, 7, 512)    1,049,088   conv5_block2_out…
(Conv2D)

conv5_block3_1_bn     (None, 7, 7, 512)        2,048   conv5_block3_1_c…
(BatchNormalizatio…

conv5_block3_1_relu   (None, 7, 7, 512)            0   conv5_block3_1_b…
(Activation)

conv5_block3_2_conv   (None, 7, 7, 512)    2,359,808   conv5_block3_1_r…
(Conv2D)

conv5_block3_2_bn     (None, 7, 7, 512)        2,048   conv5_block3_2_c…
(BatchNormalizatio…

conv5_block3_2_relu   (None, 7, 7, 512)            0   conv5_block3_2_b…
(Activation)

conv5_block3_3_conv   (None, 7, 7,         1,050,624   conv5_block3_2_r…
```

```
 (Conv2D)              2048)

 conv5_block3_3_bn     (None, 7, 7,            8,192   conv5_block3_3_c…
 (BatchNormalizatio…   2048)

 conv5_block3_add      (None, 7, 7,                0   conv5_block2_out…
 (Add)                 2048)                           conv5_block3_3_b…

 conv5_block3_out      (None, 7, 7,                0   conv5_block3_add…
 (Activation)          2048)

 reshape_2 (Reshape)   (None, 7, 14336)            0   conv5_block3_out…

 gru_features (GRU)    (None, 256)        11,208,192   reshape_2[0][0]

 artist_output_gru     (None, 24)              6,168   gru_features[0][…
 (Dense)

 genre_output_gru      (None, 10)              2,570   gru_features[0][…
 (Dense)

 style_output_gru      (None, 16)              4,112   gru_features[0][…
 (Dense)


 Total params: 34,808,754 (132.78 MB)


 Trainable params: 11,221,042 (42.80 MB)


 Non-trainable params: 23,587,712 (89.98 MB)

None
```

[ ]:
```python
from tensorflow.keras.metrics import F1Score

metrics ={
    "artist_output_gru": ["accuracy"],
    "genre_output_gru": ["accuracy"],
    "style_output_gru": ["accuracy"]
}

optimizer = tf.keras.optimizers.Adam(learning_rate=1e-4)

gru_crnn_model.compile(
    optimizer=optimizer,
```

```
    loss={
        'artist_output_gru': 'sparse_categorical_crossentropy',
        'genre_output_gru': 'sparse_categorical_crossentropy',
        'style_output_gru': 'sparse_categorical_crossentropy'
    },
    metrics = metrics
)

print("Model compiled successfully!")
```

```
Model compiled successfully!
```

[ ]:
```
history_gru_crnn = gru_crnn_model.fit(
    train_data_tf,
    validation_data=val_data_tf,
    epochs=10
)
```

```
Epoch 1/10
25/25              7s 255ms/step -
artist_output_crnn_accuracy: 0.2799 - artist_output_crnn_loss: 2.5094 -
genre_output_crnn_accuracy: 0.4178 - genre_output_crnn_loss: 1.7363 - loss:
5.9178 - style_output_crnn_accuracy: 0.4418 - style_output_crnn_loss: 1.6721 -
val_artist_output_crnn_accuracy: 0.1800 - val_artist_output_crnn_loss: 2.8745 -
val_genre_output_crnn_accuracy: 0.3700 - val_genre_output_crnn_loss: 1.9456 -
val_loss: 6.4245 - val_style_output_crnn_accuracy: 0.4300 -
val_style_output_crnn_loss: 1.7343
Epoch 2/10
25/25              7s 276ms/step -
artist_output_crnn_accuracy: 0.3017 - artist_output_crnn_loss: 2.4029 -
genre_output_crnn_accuracy: 0.3977 - genre_output_crnn_loss: 1.7590 - loss:
5.8136 - style_output_crnn_accuracy: 0.4895 - style_output_crnn_loss: 1.6517 -
val_artist_output_crnn_accuracy: 0.2000 - val_artist_output_crnn_loss: 2.8420 -
val_genre_output_crnn_accuracy: 0.3400 - val_genre_output_crnn_loss: 1.8332 -
val_loss: 6.3666 - val_style_output_crnn_accuracy: 0.4000 -
val_style_output_crnn_loss: 1.6105
Epoch 3/10
25/25              10s 383ms/step -
artist_output_crnn_accuracy: 0.3094 - artist_output_crnn_loss: 2.4058 -
genre_output_crnn_accuracy: 0.3935 - genre_output_crnn_loss: 1.7239 - loss:
5.8104 - style_output_crnn_accuracy: 0.4592 - style_output_crnn_loss: 1.6807 -
val_artist_output_crnn_accuracy: 0.1700 - val_artist_output_crnn_loss: 2.6851 -
val_genre_output_crnn_accuracy: 0.3400 - val_genre_output_crnn_loss: 1.9053 -
val_loss: 6.3625 - val_style_output_crnn_accuracy: 0.4200 -
val_style_output_crnn_loss: 1.6316
Epoch 4/10
25/25              8s 280ms/step -
artist_output_crnn_accuracy: 0.3237 - artist_output_crnn_loss: 2.3336 -
```

```
genre_output_crnn_accuracy: 0.3984 - genre_output_crnn_loss: 1.7519 - loss:
5.6568 - style_output_crnn_accuracy: 0.4999 - style_output_crnn_loss: 1.5712 -
val_artist_output_crnn_accuracy: 0.1800 - val_artist_output_crnn_loss: 2.7250 -
val_genre_output_crnn_accuracy: 0.3500 - val_genre_output_crnn_loss: 2.0155 -
val_loss: 6.4042 - val_style_output_crnn_accuracy: 0.4200 -
val_style_output_crnn_loss: 1.6558
Epoch 5/10
25/25              6s 249ms/step -
artist_output_crnn_accuracy: 0.3257 - artist_output_crnn_loss: 2.3479 -
genre_output_crnn_accuracy: 0.4144 - genre_output_crnn_loss: 1.7035 - loss:
5.6259 - style_output_crnn_accuracy: 0.4992 - style_output_crnn_loss: 1.5744 -
val_artist_output_crnn_accuracy: 0.2000 - val_artist_output_crnn_loss: 2.7919 -
val_genre_output_crnn_accuracy: 0.3600 - val_genre_output_crnn_loss: 1.7839 -
val_loss: 6.2327 - val_style_output_crnn_accuracy: 0.4300 -
val_style_output_crnn_loss: 1.6265
Epoch 6/10
25/25              9s 349ms/step -
artist_output_crnn_accuracy: 0.3143 - artist_output_crnn_loss: 2.3643 -
genre_output_crnn_accuracy: 0.4531 - genre_output_crnn_loss: 1.6346 - loss:
5.5592 - style_output_crnn_accuracy: 0.5042 - style_output_crnn_loss: 1.5603 -
val_artist_output_crnn_accuracy: 0.1700 - val_artist_output_crnn_loss: 2.7140 -
val_genre_output_crnn_accuracy: 0.3300 - val_genre_output_crnn_loss: 1.9347 -
val_loss: 6.3598 - val_style_output_crnn_accuracy: 0.4600 -
val_style_output_crnn_loss: 1.6157
Epoch 7/10
25/25              6s 242ms/step -
artist_output_crnn_accuracy: 0.3145 - artist_output_crnn_loss: 2.3478 -
genre_output_crnn_accuracy: 0.4205 - genre_output_crnn_loss: 1.6763 - loss:
5.6407 - style_output_crnn_accuracy: 0.4750 - style_output_crnn_loss: 1.6166 -
val_artist_output_crnn_accuracy: 0.1700 - val_artist_output_crnn_loss: 2.8218 -
val_genre_output_crnn_accuracy: 0.3300 - val_genre_output_crnn_loss: 1.9559 -
val_loss: 6.5656 - val_style_output_crnn_accuracy: 0.4100 -
val_style_output_crnn_loss: 1.7369
Epoch 8/10
25/25              10s 241ms/step -
artist_output_crnn_accuracy: 0.3235 - artist_output_crnn_loss: 2.3021 -
genre_output_crnn_accuracy: 0.4751 - genre_output_crnn_loss: 1.6005 - loss:
5.4370 - style_output_crnn_accuracy: 0.5187 - style_output_crnn_loss: 1.5343 -
val_artist_output_crnn_accuracy: 0.2200 - val_artist_output_crnn_loss: 2.7934 -
val_genre_output_crnn_accuracy: 0.3000 - val_genre_output_crnn_loss: 1.9277 -
val_loss: 6.3198 - val_style_output_crnn_accuracy: 0.4200 -
val_style_output_crnn_loss: 1.6294
Epoch 9/10
25/25              7s 281ms/step -
artist_output_crnn_accuracy: 0.3362 - artist_output_crnn_loss: 2.2734 -
genre_output_crnn_accuracy: 0.4429 - genre_output_crnn_loss: 1.6219 - loss:
5.4215 - style_output_crnn_accuracy: 0.5153 - style_output_crnn_loss: 1.5262 -
val_artist_output_crnn_accuracy: 0.2100 - val_artist_output_crnn_loss: 2.8020 -
```

```
val_genre_output_crnn_accuracy: 0.3200 - val_genre_output_crnn_loss: 1.9906 -
val_loss: 6.3250 - val_style_output_crnn_accuracy: 0.4500 -
val_style_output_crnn_loss: 1.6583
Epoch 10/10
25/25              7s 267ms/step -
artist_output_crnn_accuracy: 0.3410 - artist_output_crnn_loss: 2.2246 -
genre_output_crnn_accuracy: 0.4267 - genre_output_crnn_loss: 1.6574 - loss:
5.4419 - style_output_crnn_accuracy: 0.4980 - style_output_crnn_loss: 1.5599 -
val_artist_output_crnn_accuracy: 0.1600 - val_artist_output_crnn_loss: 2.6982 -
val_genre_output_crnn_accuracy: 0.3600 - val_genre_output_crnn_loss: 1.8530 -
val_loss: 6.2868 - val_style_output_crnn_accuracy: 0.4200 -
val_style_output_crnn_loss: 1.7426
```

3.3 Classification Metrics for CRNN

3.3.1 Results for CRNN using LSTM

[ ]:
```
results_crnn = crnn_model.evaluate(test_data_tf, return_dict=True)
for k, v in results_crnn.items():
    print(f"{k}: {v:.4f}")
```

```
4/4              1s 139ms/step -
artist_output_crnn_accuracy: 0.1694 - artist_output_crnn_loss: 2.9302 -
genre_output_crnn_accuracy: 0.2467 - genre_output_crnn_loss: 2.0653 - loss:
6.9750 - style_output_crnn_accuracy: 0.3815 - style_output_crnn_loss: 2.0351
artist_output_crnn_accuracy: 0.1500
artist_output_crnn_loss: 3.0382
genre_output_crnn_accuracy: 0.2600
genre_output_crnn_loss: 1.9707
loss: 7.0558
style_output_crnn_accuracy: 0.3600
style_output_crnn_loss: 2.1860
```

3.3.2 Results for CRNN using GRU

[ ]:
```
results_gru_crnn = gru_crnn_model.evaluate(test_data_tf, return_dict=True)
for k, v in results_gru_crnn.items():
    print(f"{k}: {v:.4f}")
```

```
4/4              6s 138ms/step -
artist_output_gru_accuracy: 0.0460 - artist_output_gru_loss: 3.8009 -
genre_output_gru_accuracy: 0.0431 - genre_output_gru_loss: 2.6196 - loss: 9.1991
- style_output_gru_accuracy: 0.0368 - style_output_gru_loss: 2.6958
artist_output_gru_accuracy: 0.0500
artist_output_gru_loss: 3.7040
genre_output_gru_accuracy: 0.0400
genre_output_gru_loss: 2.4854
loss: 9.1025
```

```
style_output_gru_accuracy: 0.0400
style_output_gru_loss: 2.7058
```

3.3.3 F1 Score for CRNN using LSTM

[ ]:
```
# calculating f1_scores for crnn model
y_pred_crnn = crnn_model.predict(test_data_tf)
y_pred_crnn_classes = [np.argmax(pred, axis=1) for pred in y_pred_crnn]

f1_scores_crnn = [
    f1_score(y_true_np[i], y_pred_crnn_classes[i], average='macro')
    for i in range(3)
]

overall_f1_crnn = sum(f1_scores_crnn) / len(f1_scores_crnn)
print("Overall Macro F1 for CRNN : ", overall_f1_crnn)
```

```
4/4              6s 681ms/step
Overall Macro F1 for CRNN :  0.08769100328303363
```

3.3.4 F1 socre for CRNN using GRU

[ ]:
```
# calculating f1_scores for crnn_gru model
y_pred__gru_crnn = gru_crnn_model.predict(test_data_tf)
y_pred_gru_crnn_classes = [np.argmax(pred, axis=1) for pred in y_pred__gru_crnn]

f1_scores_gru_crnn = [
    f1_score(y_true_np[i], y_pred_gru_crnn_classes[i], average='macro')
    for i in range(3)
]

overall_f1_gru_crnn = sum(f1_scores_gru_crnn) / len(f1_scores_gru_crnn)
print("Overall Macro F1 for GRU CRNN : ", overall_f1_gru_crnn)
```

```
4/4              10s 2s/step
Overall Macro F1 for GRU CRNN :  0.006152138637923158
```

3.3.5 LSTM vs GRU

- In theory, GRU should surpass LSTM in this case because of smaller dataset size
- Trains faster than LSTM (which is true here)
- But viewig the F1 scores and accuracy and the loss values, it's evident that the LSTM performed much better than GRU (0.087 vs 0.006 as in F1 scores)

So, undoubtedly, i proceeded with using LSTM for my recurrent layers

3.3.6 Predicted Class Distribution for CRNN using LSTM

[ ]:
```
for i in range(3):
    print(f"\nOutput {i+1}")
    print("True distribution:", np.bincount(y_true_np[i]))
    print("Pred distribution:", np.bincount(y_pred_classes[i]))
```

```
Output 1
True distribution: [ 1  4  5  4 10  3  1  3  3  0  1  3  3  4  9  2  7  7  5  3
5  3  8  6]
Pred distribution: [ 0  0  0  0  0  0  0  0  0  0  0  0  0  0 42  0  0  0  0  0
0  0 58]

Output 2
True distribution: [ 9 14  2 27  2 20  6  4  4 12]
Pred distribution: [ 0  0  0 97  0  3]

Output 3
True distribution: [ 0  0  8  5  5  3  1 32  2  7  0 14 10  4  9]
Pred distribution: [ 0  0  0  0  0  0  0 95  0  0  0  0  5]
```

3.3.7 Classification report for CRNN using LSTM

[ ]:
```
from sklearn.metrics import classification_report

for i in range(3):
    print(f"\nOutput {i+1}")
    print(classification_report(y_true_np[i], y_pred_crnn_classes[i]))
```

```
Output 1
              precision    recall  f1-score   support

           0       0.00      0.00      0.00         1
           1       0.00      0.00      0.00         4
           2       0.00      0.00      0.00         5
           3       0.00      0.00      0.00         4
           4       0.29      0.60      0.39        10
           5       0.00      0.00      0.00         3
           6       0.00      0.00      0.00         1
           7       0.00      0.00      0.00         3
           8       0.00      0.00      0.00         3
           9       0.00      0.00      0.00         0
          10       0.00      0.00      0.00         1
          11       0.00      0.00      0.00         3
          12       0.00      0.00      0.00         3
          13       0.00      0.00      0.00         4
```

14	0.50	0.33	0.40	9
15	0.00	0.00	0.00	2
16	0.00	0.00	0.00	7
17	0.00	0.00	0.00	7
18	0.00	0.00	0.00	5
19	0.00	0.00	0.00	3
20	0.00	0.00	0.00	5
21	0.00	0.00	0.00	3
22	0.10	0.75	0.17	8
23	0.00	0.00	0.00	6
accuracy			0.15	100
macro avg	0.04	0.07	0.04	100
weighted avg	0.08	0.15	0.09	100

Output 2

	precision	recall	f1-score	support
0	0.00	0.00	0.00	9
1	0.33	0.07	0.12	14
2	0.50	0.50	0.50	2
3	0.35	0.63	0.45	27
4	0.00	0.00	0.00	2
5	1.00	0.05	0.10	20
6	0.00	0.00	0.00	6
7	0.00	0.00	0.00	4
8	0.00	0.00	0.00	4
9	0.14	0.50	0.21	12
accuracy			0.26	100
macro avg	0.23	0.18	0.14	100
weighted avg	0.37	0.26	0.19	100

Output 3

	precision	recall	f1-score	support
2	0.00	0.00	0.00	8
3	0.00	0.00	0.00	5
4	0.00	0.00	0.00	5
5	0.00	0.00	0.00	3
6	0.00	0.00	0.00	1
7	0.36	0.97	0.52	32
8	0.00	0.00	0.00	2
9	0.00	0.00	0.00	7
11	0.83	0.36	0.50	14
12	0.00	0.00	0.00	10

```
          13       0.00      0.00      0.00         4
          14       0.00      0.00      0.00         9

    accuracy                           0.36       100
   macro avg       0.10      0.11      0.09       100
weighted avg       0.23      0.36      0.24       100

/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565:
UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels
with no predicted samples. Use `zero_division` parameter to control this
behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565:
UndefinedMetricWarning: Recall is ill-defined and being set to 0.0 in labels
with no true samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565:
UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels
with no predicted samples. Use `zero_division` parameter to control this
behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565:
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with no true samples. Use `zero_division` parameter to control this behavior.
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/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565:
UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels
with no predicted samples. Use `zero_division` parameter to control this
```

```
behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565:
UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels
with no predicted samples. Use `zero_division` parameter to control this
behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565:
UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels
with no predicted samples. Use `zero_division` parameter to control this
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  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
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UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels
with no predicted samples. Use `zero_division` parameter to control this
behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```
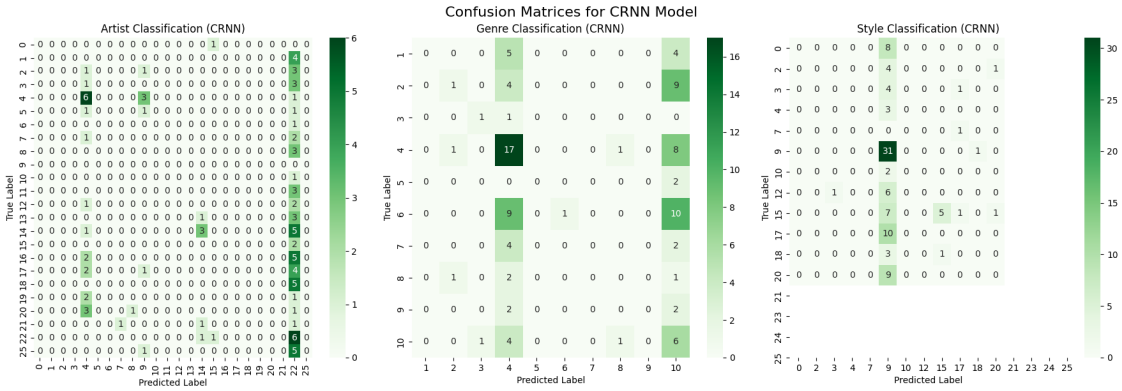

3.3.8 Confusion matrix for CRNN using LSTM

[ ]:
```
output_titles = ['Artist Classification (CRNN)', 'Genre Classification (CRNN)',␣
↪'Style Classification (CRNN)']

plt.figure(figsize=(18, 6))

for i in range(3):
    cm_crnn = confusion_matrix(y_true_np[i], y_pred_crnn_classes[i])

    current_class_names = all_class_names[['artist', 'genre', 'style'][i]]

    plt.subplot(1, 3, i + 1)
    sns.heatmap(cm_crnn, annot=True, fmt='d', cmap='Greens',
                xticklabels=current_class_names,␣
↪yticklabels=current_class_names)
    plt.title(output_titles[i])
    plt.xlabel('Predicted Label')
    plt.ylabel('True Label')

plt.tight_layout()
plt.suptitle('Confusion Matrices for CRNN Model', y=1.02, fontsize=16)
plt.show()
```



3.3.9 Outlier Detection for CRNN using LSTM

[ ]:
```
import numpy as np
import matplotlib.pyplot as plt

all_images = []
all_labels = []
for images_batch, labels_batch in test_data_tf:
    all_images.append(images_batch.numpy())
    all_labels.append(labels_batch[0].numpy())
all_images = np.concatenate(all_images, axis=0)
all_labels = np.concatenate(all_labels, axis=0)

artist_preds_probs_crnn = crnn_model.predict(test_data_tf, verbose=0)[0]

misclassified_indices_crnn = np.where(y_true_np[0] != y_pred_crnn_classes[0])[0]

print(f"Total misclassified samples for Artist (CRNN):␣
 ↪{len(misclassified_indices_crnn)}")

num_to_display = min(5, len(misclassified_indices_crnn))

if num_to_display > 0:
    np.random.shuffle(misclassified_indices_crnn)
    display_indices = misclassified_indices_crnn[:num_to_display]

    plt.figure(figsize=(15, 5))
    for i, idx in enumerate(display_indices):
        true_label = y_true_np[0][idx]
        predicted_label = y_pred_crnn_classes[0][idx]
        confidence = artist_preds_probs_crnn[idx, predicted_label]

        plt.subplot(1, num_to_display, i + 1)
```

```
        plt.imshow(all_images[idx])
        plt.axis('off')
        plt.title(f"True: {true_label}\nPred: {predicted_label}\nConf:␣
↪{confidence:.2f}", fontsize=10)
    plt.tight_layout()
    plt.suptitle('CRNN Random Misclassifications (Artist)', y=1.05, fontsize=16)
    plt.show()
else:
    print("No misclassified samples found to display for CRNN.")
```

```
Total misclassified samples for Artist (CRNN): 85
```

3.3.10 Analysis of CRNN Model :

Metrics similar to CNN were used for performance analysis of CRNN model.

- **Overall Macro F1** : CRNN slightly has higher overall macro F1 score (0.083) vs (0.034) for CNN, this suggests that the LSTM component have certainly captured some useful context.
- **Class Distribution and Classification Report** : CRNN also suffered from severe class imbalance, with prediction biased towards dominant classes.

Conclusion for CRNN : While CRNN showed a significant improvement in F1 score, accuracy and loss over the CNN only architecture, the problems faced by both the models were the same : - Small dataset - Class imbalance

These could be solved by : - Data augmentation - Adding class weights to minority classes while training.

3.4 Comparative Analysis of CNN vs. CRNN

Comparing the key performance metrics for three of the models.

Metric	CNN (ResNet50)	CRNN (ResNet50 + LSTM)	CRNN (ResNet50 + GRU)
Overall Macro F1	0.034	0.083	0.006
Artist Output Accuracy	0.120	0.170	0.050
Genre Output Accuracy	0.230	0.290	0.040

Metric	CNN (ResNet50)	CRNN (ResNet50 + LSTM)	CRNN (ResNet50 + GRU)
Style Output Accuracy	0.340	0.380	0.050

Summary of Comparison:

- The **CRNN model with LSTM showed a modest improvement** across all metrics (accuracy and overall Macro F1) compared to the CNN. This suggests that the recurrent component, even with a small dataset, might offer a slight advantage in processing the spatial features for classification.
- The GRU model however, performed even worse than the only CNN model, which was surprising to note here and was not expected.
- However, **both (LSTM and CNN) models still exhibit very low performance**, especially as indicated by the Macro F1-scores. This strongly points to underlying data limitations rather than architectural shortcomings as the primary bottleneck.
- The **class imbalance problem** was pervasive in both models, leading to biased predictions towards majority classes.

3.4.1 Outlier Detection

I initially attempeted to bring out : - Low-confidence predictions - High-confidence misclassifications (model was incorrect but highly confident (threshold > 0.5))

Since, both the attempts did not yield useful results, I proceeded with identifying random misclassified samples.

This allowed to understand the model's failure to generalize due to limited data.

3.5 Overall Conclusion

This notebook successfully demonstrated the implementation and evaluation of CNN and CRNN models for multi-label image classification on a constrained dataset, effectively utilizing transfer learning with ResNet50 and explored the benefits of adding recurrent layers.

Addressed all aspects of the task:

- **Model Architectures:** Defined and implemented both CNN and CRNN.
- **Evaluation Metrics:** Critically selected and applied appropriate metrics (Macro F1, Classification Report, Confusion Matrices) suitable for imbalanced data, not relying on simple accuracy.
- **Outlier Detection:** Visualized misclassified samples as a form of outliers, providing insights of the model's failures at certain images.

Primary Problems I faced:

1. **Limited Dataset Size:** Due to computation constraints (using google colab) i was restricted to using small dataset (wikiart-tiny with just 1000 images) which is insufficient for training robust models, even with pretrained backbones and using transfer learning leading to poor feature extraction and generalization.

2. **Severe Class Imbalance:** The uneven distribution of samples across artists, genres, and styles heavily biased model predictions, resulting in extremely low performance for minority classes.

Possible solutions:

To significantly improve model performance, overcoming the above limitations, :

- **Data Augmentation:** Applying transformations and increasing the dataset size can help for improvement.
- **Class Weighting:** Assigning higher weights to minority classes in the loss function can make the model to learn those classes better.
- **More Data:** A better dataset with balanced distribution and enough size.
- **Fine-tuning:** Lowering the learning rate of CNN gave me a little better result, so unfreezing and fine-tuning some of the top layers of resnet50 with small learning rate and help the model to learn more specific features.

This detailed analysis provides a strong starting point with possible problems and solutions that can help architect more better models and much better data handling for achieving bettter performance on such challenging multi-label classification tasks.